



CAREER APEX CRM

ENTERPRISE SOFTWARE ENGINEERING SPECIFICATION

Full Software Development Lifecycle (SDLC) Master Blueprint

Complete Business Architecture, User Scoping, System Topology, Database DDL, REST Contracts, 14 Screen Functional Manual & QA Acceptance Test Plan

- VERSION 2.0.0
- STUDENT JOB-SEEKER DOMAIN
- PLACEMENT FEE LEDGER
- PRODUCTION PRE-FLIGHT

DOMAIN FOCUS	DOCUMENT SCOPE
Candidate Job Placement & Career Counseling CRM	BRS, URS, PRD, Screen Guide, HLD, LLD, DB, API, Tech Stack, Test Plan
TARGET AUDIENCE	STATUS & DATE
Engineering Leads, Enterprise Architects, Product Managers, QA Engineers	Certified Master Release — September 2026

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This consolidated master document integrates the complete lifecycle specifications for the Career Apex CRM platform. Each section represents a standalone, rigorously audited technical artifact.

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Career Apex CRM — Software Development Lifecycle (SDLC) Documentation Suite

Master Document Index & Navigation Guide

Project Overview

- **Project Name:** Career Apex Sales CRM (Student Placement & Career Counseling Sales CRM)
- **Domain:** Higher Education, Job Seeking, Career Placement & Vocational Counseling
- **System Version:** 2.0 (Prototype to Full-Stack Enterprise Blueprint)
- **Author:** Antigravity Engineering Architecture Team
- **Date:** September 2026

Purpose of this Documentation Suite

This directory contains the formal, end-to-end software development lifecycle (SDLC) specifications for Career Apex CRM. Each document serves a dedicated phase in the engineering lifecycle—from executive business requirements, product definition, and user role scoping, to high-level architecture, low-level component design, relational schema, API contracts, screen-by-screen functional blueprints, technology stacks, and quality assurance test plans.

Documentation Suite Map

Document Code	Document Title	Target Audience	Primary Focus
[01_BRS]	Business Requirements Specification	Business Owners, Sales Leadership, Stakeholders	Business problem statement, ROI goals, candidate lifecycle, placement fee monetization, operational KPIs.
[02_URS]	User Requirements & Role Hierarchy	HR, Project Managers, Security Architects	User personas (Admin, Manager, Counselor), organizational hierarchy, scope of data access, RBAC matrix.
[03_PRD]	Product Requirements Document	Product Managers, UI/UX Leads, Tech Leads	Functional requirements (FRs), non-functional requirements (NFRs), system rules, edge cases, acceptance criteria.
[04_SCREEN]	Screen-by-Screen Functional Specification	Frontend Developers, QA Engineers, Designers	Exhaustive deep-dive into all 14 screens, UI layout, components, buttons, filters, modals, workflows, and state changes.
[05_HLD]	High-Level Design Document	Enterprise Architects, Lead Developers	System topology, C4 architectural models, data flow sequences, security, session management, and scalability.
[06_LLD]	Low-Level Design Document	Full-Stack Developers, Module Owners	Component class diagrams, state machines (Pipeline & Payments), algorithmic data processing, and validation rules.
[07_DB]	Database Design & Data Dictionary	Database Administrators, Backend Engineers	Relational ERD schema, PostgreSQL DDL statements, field definitions, foreign keys, indexes, and integrity rules.
[08_API]	RESTful API Specification	Backend & Integration Developers	REST API contracts, endpoints, query parameters, request/response JSON payloads, status codes, and error models.
[09_STACK]	Tech Stack & Migration Roadmap	DevOps, Tech Leads, Engineering Managers	Production technology stack recommendations, infrastructure setup, and 6-phase migration roadmap from prototype.
[10_TEST]	Test Plan & Acceptance Test Cases	QA Leads, Automation Engineers, Product Owners	Comprehensive testing strategy, test scenarios, manual/automated test cases, regression checklists, and test data sets.

Reading Recommendations by Role

- **For Business Executives & Product Managers:** Start with `01_BRS` , `02_URS` , `03_PRD` , and `04_SCREEN` .
- **For Technical Architects & Tech Leads:** Review `05_HLD` , `06_LLD` , `07_DB` , and `09_STACK` .
- **For Frontend & Backend Developers:** Read `03_PRD` , `04_SCREEN` , `06_LLD` , `07_DB` , and `08_API` .
- **For Quality Assurance (QA) & Testers:** Focus on `03_PRD` , `04_SCREEN` , `08_API` , and `10_TEST` .

Business Requirements Specification (BRS)

`01_BRS_BUSINESS_REQUIREMENTS_SPECIFICATION.md`

Business Requirements Specification (BRS)

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** BRS-APEX-01
- **Version:** 2.0
- **Status:** Approved
- **Domain:** Student Career Placement, EdTech Sales, Job Counseling Agency

1. Business Context & Problem Statement

1.1 Industry Background

Educational placement institutes, job training academies, and staffing/career counseling consultancies operate with a unique operational model:

1. They source thousands of aspiring student job seekers across engineering colleges, universities, and polytechnics.
2. They guide them through career counseling, technical skills assessment, and corporate interview readiness.
3. They connect qualified students with corporate recruiters and place them into technical, operational, and professional roles.
4. **Monetization Model:** Candidates agree to pay a **Placement Service Fee** (ranging from ₹30,000 to ₹70,000) either upfront upon registration or in structured milestones (e.g., training milestone, interview clearance milestone, joining/placement milestone).

1.2 Critical Flaws in Traditional B2B CRMs

Traditional CRMs such as Salesforce, HubSpot, or Zoho are built around corporate B2B transactions:

- **Corporate Account Anchor:** Every contact must belong to an "Account" or "Company." For student recruitment, students are independent job seekers and have no company account.
- **Deal-Centric Pipelines:** Standard deals track vendor procurement stages instead of student educational qualifications, passing years (YOP), skills, and interview readiness.

- **Payment Inflexibility:** Traditional CRMs do not natively track student installment milestones, UTR references, receipts, and overdue collections against candidates.
- **Sub-optimal Tabular Views:** Counselors handling hundreds of cold calls daily need spreadsheet-speed data entry, not complex drag-and-drop Kanban boards.

1.3 Proposed Business Solution

Career Apex CRM is engineered specifically for candidate placement counseling:

- **Student Candidate as First-Class Entity:** Comprehensive capture of degree, college, passing year (2022–2026), skills, target roles, and location preferences.
- **Excel Spreadsheet-Style Interface:** Enables counselors to review and progress hundreds of leads daily with instant inline calling, WhatsApp, and status updates.
- **Native Placement Fee Management:** Configurable milestone fee plans (Upfront, 2, 3, or 4 installments) with payment recording, transaction references, balance calculation, and receipt issuance.
- **Multi-Level Intelligence:** Real-time visibility into counselor conversion rates, fee realization, and acquisition channel efficiency.

2. Business Objectives & ROI Metrics

Strategic Objective	Baseline Metric	Target Metric	CRM Enabling Capability
Lead Outreach Speed	48–72 hours post-inquiry	< 4 hours (90% within 24h)	Cold Calling pipeline sheet, 1-click WhatsApp & Call integration
Placement Fee Realization	62% of booked fees collected	> 88% realization	Installment schedule tracking, overdue alerts, payment receipt generation
Counselor Productivity	~20 interactions/day	45+ candidate touches/day	Dense Excel view, inline call outcome logging, rapid follow-up scheduler
Lead Ingestion Efficiency	2 days manual data entry	< 3 minutes bulk upload	SheetJS Excel/CSV import wizard with duplicate phone detection
Drop-off Auditability	Unknown lost reasons	100% audited stage changes	Mandatory transition reason logging on every stage change

3. Business Scope

3.1 In-Scope Capabilities

1. **Candidate Onboarding:** Individual registration and mass Excel/CSV spreadsheet upload.
2. **Contact Deduplication:** Strict phone number uniqueness validation across primary and alternate mobile numbers.
3. **Pipeline Staging:** 10 sequential placement stages:

- *Cold Calling (Raw Data) → New Lead → Contacted → Interested → Prospect → Follow-up → Negotiation → Converted (Placed) → Not Interested → Lost.*

- 4. Placement Fee Engine:** Custom fee assignment, installment generation, transaction logging (UPI, Net Banking, Card, Cash, Cheque), real-time balance computation, and printable receipt generation.
- 5. Counseling Touchpoints:** Telephony call outcome logging, WhatsApp templated messaging, email communication.
- 6. Task Execution:** Scheduled follow-ups categorized into Overdue, Today, and Upcoming.
- 7. Role Governance:** Strict role segregation for Super Administrator, Sales Manager, and Career Counselor.
- 8. Universal Reporting:** Multi-dimensional filtering across dates, counselors, managers, stages, and fee statuses.

3.2 Out-of-Scope (Deferred to Future Releases)

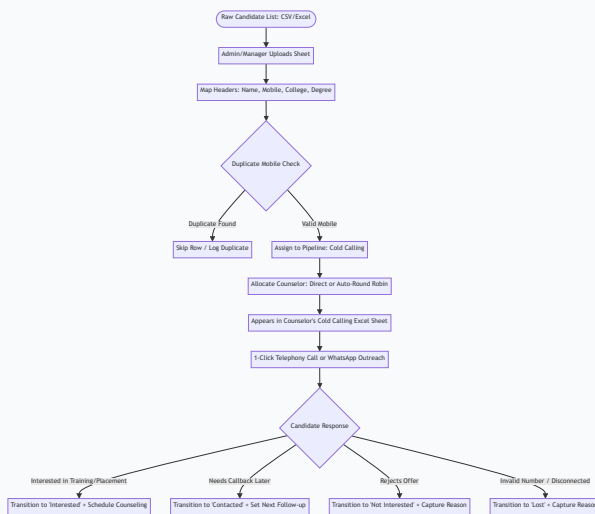
- Automated SMS gateway integration (e.g. Twilio/Gupshup) — scheduled for Phase 4.
- Automated WhatsApp Cloud API bot dispatch — scheduled for Phase 4.
- Direct Cloud PBX VoIP dialing (currently opens system tel: protocol) — scheduled for Phase 5.
- Online payment gateway integration (Razorpay/Stripe checkout links) — cash/manual bank transfer tracking supported in current scope.

4. Business Actors & Stakeholders

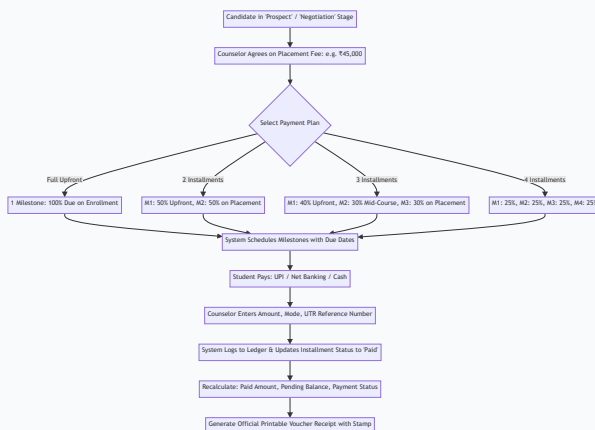
Stakeholder Role	Business Responsibility	Primary System Interest
Super Administrator (Executive)	Company director, operations head, or CEO. Overall business health, revenue targets, and compliance.	Executive dashboards, booked vs collected placement revenue, organization-wide conversion rates.
Sales / Center Manager	Team leader managing 5–15 career counselors. Lead allocation, quota tracking, and team coaching.	Team pipeline spreadsheet, lead reassignment tools, counselor productivity leaderboards.
Career Counselor (Sales Rep)	Front-line counselor conducting counseling calls, pitching placement tracks, scheduling interviews, and collecting fees.	Daily action queue, 1-click calling/WhatsApp, student dossier, recording payments.
Student Candidate (External Subject)	College student or job seeker looking for corporate placement.	Timely counseling, clear milestone fees, official payment receipts.

5. End-to-End Business Workflows

5.1 Raw Database Ingestion & Cold Calling Lifecycle



5.2 Placement Fee Agreement & Cash Realization Workflow



6. Business Rules (BR)

- **BR-01 (Mandatory Unique Identifier):** Every registered student candidate must have a unique 10-digit primary mobile number. Duplicate entries with the same mobile or matching alternate WhatsApp numbers must be rejected.
- **BR-02 (Mandatory Stage Transition Reason):** A counselor or manager cannot change a candidate's pipeline stage without supplying an explicit business reason (e.g. "Completed technical round," "Declined due to location preference").

- **BR-03 (Placement Fee Invariance):** Total placement fee must equal the sum of cash collected (`paidAmount`) plus outstanding balance (`pendingAmount`). Overpayments are disallowed.
 - **BR-04 (Installment Due Date Enforcement):** Any pending milestone with a due date earlier than the current business date must be automatically flagged as `Overdue` in red accents across all dashboards and reports.
 - **BR-05 (Receipt Numbering):** Every approved transaction must generate a sequential voucher reference code (`REC-XXXXX`) that cannot be modified once committed.
 - **BR-06 (Data Scoping):** Career counselors can only view and act on students assigned to their user ID. Sales managers can only view students assigned to counselors in their reporting team. Super administrators have global visibility.
-

7. Assumptions & Dependencies

- Counselors have internet connectivity and desktop/laptop browsers supporting HTML5 standards.
- Candidate mobile numbers belong to the standard Indian telecommunications numbering plan (+91 10-digit format).
- Initial system deployment uses local storage persistence with zero external cloud dependencies for seamless, immediate demonstration.

User Requirements & Role Hierarchy (URS)

02_URS_USER_ROLES_AND_PERMISSIONS.md

User Requirements & Role Hierarchy Specification (URS)

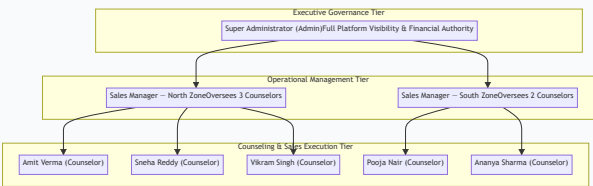
Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** URS-APEX-02
- **Version:** 2.0
- **Status:** Approved
- **Audience:** Product Managers, System Architects, Security Engineers, QA Teams

1. Purpose & User Archetypes

The Career Apex CRM enforces a strict three-tier organizational hierarchy. Each role corresponds to specific operational duties, data visibility boundaries, and transactional permissions within the career counseling and student placement lifecycle.



2. Detailed User Personas & Jobs-to-be-Done (JTBD)

2.1 Persona 1: Super Administrator

- **Representative Profile:** Rajesh Sharma (ID: `USR-001`), Director of Placement Operations.
- **Primary Goal:** Maximize student intake, ensure team compliance, audit fee realization, and eliminate bottlenecks in placement hiring.

- **Core Frustrations:** Lack of centralized visibility into booked vs collected fees; duplicate candidate records; counselors hoarding uncontacted leads.
- **Jobs-to-be-Done (JTBD):**
 - "When reviewing monthly operations, I want to see total placement revenue contracted, cash realized, and overdue balances across all centers."
 - "When onboarding bulk college lists, I want to upload 5,000 students at once and automatically distribute them to counselors without duplicate mobile numbers."
 - "When a counselor resigns or underperforms, I want to reassign their entire candidate portfolio to another counselor with one click."

2.2 Persona 2: Sales Manager

- **Representative Profile:** Priya Patel (ID: `USR-002`), Regional Sales Manager (North Zone).
- **Primary Goal:** Drive team counseling targets, track daily calls/follow-ups, and monitor installment collections across direct reporting counselors.
- **Core Frustrations:** Counselors forgetting to follow up with warm prospects; milestone payments lapsing into overdue status without notice.
- **Jobs-to-be-Done (JTBD):**
 - "Every morning, I want to view my team command center to see today's pending follow-ups and overdue dues under my reporting counselors."
 - "When raw cold-calling data arrives, I want to allocate specific batches to my team members based on their target quotas."
 - "When reviewing weekly performance, I want to benchmark counselors by lead-to-conversion win rates and cash collected."

2.3 Persona 3: Career Counselor / Sales Representative

- **Representative Profile:** Amit Verma (ID: `USR-004`), Senior Career Counselor.
- **Primary Goal:** Rapidly connect with student job seekers, explain training and placement benefits, schedule follow-ups, agree on placement fees, and collect milestone payments.
- **Core Frustrations:** Complex CRMs that require 10 clicks to log a call; having to switch to WhatsApp Web manually; losing track of student installment due dates.
- **Jobs-to-be-Done (JTBD):**
 - "When I log in, I want an immediate action queue showing exactly which students I need to call today and which follow-ups are overdue."
 - "When viewing my Cold Calling sheet, I want to click one button to trigger a call or open a pre-filled WhatsApp message without navigating away."
 - "When a student pays an installment via UPI, I want to enter the UTR reference number and instantly generate a formatted, printable receipt."

3. Data Scoping & Visibility Boundaries

The application enforces automatic data scoping at the data layer (`Customers.getScopedCustomers(user)`):

Data Domain	Super Administrator Scope	Sales Manager Scope	Career Counselor Scope
Student Profiles	Global: All registered students in the organization.	Team-Scoped: Students assigned to counselors reporting to this manager.	Personal: Students directly assigned to this counselor's user ID.
Pipeline Views	Views all students across all stages.	Views team students across all stages.	Views personal students across all stages.
Placement Fees & Payments	Audits all revenue, collections, and overdue dues across all centers.	Monitors booked vs collected fees for own reporting team.	Tracks booked fees, collections, and dues for personal candidates.
Call Logs & Follow-ups	Global audit trail of all company communications.	All calls and scheduled follow-ups created by team counselors.	Personal scheduled calls and assigned follow-ups.
Reports & Leaderboards	Full access to all 4 reports with global multi-filters.	Team performance, team revenue, and team stage velocity.	View personal conversion benchmarks and revenue targets.
User & Team Management	Full CRUD on all managers and counselors.	View reporting team members; request reassignment.	View own user profile.

4. Role-Based Access Control (RBAC) Matrix

System Action / Feature	Admin (admin)	Sales Manager (manager)	Career Counselor (saLes)
Access Login & Session Switching	Full	Full	Full
View Executive Dashboard	Global	Team Command Center	Personal Workspace
Register New Student Candidate	Yes	Yes	Yes
Edit Student Profile (Academic/Personal)	Yes	Yes	Yes
Edit Student Placement Fee & Plan	Yes	Yes	View / Update via Payment Record
Delete Student Record	Yes (Admin Confirmation)	No	No
Move Pipeline Stage	Yes (With reason)	Yes (With reason)	Yes (With reason)
Bulk Import Leads (Excel/CSV)	Yes (Global assign)	Yes (Team assign)	View-only
Reassign Student Counselor	Yes (Any user)	Yes (Within team)	No
Record Milestone Fee Payment	Yes	Yes	Yes (Assigned candidates)
Print Official Payment Receipt	Yes	Yes	Yes
Log Telephony Call Outcome	Yes	Yes	Yes
Trigger WhatsApp Direct Outreach	Yes	Yes	Yes
Schedule / Complete Follow-up Task	Yes	Yes	Yes
Export Student Directory to CSV	Yes	Yes	Yes (Assigned only)
Export Revenue & Team Reports to CSV	Yes	Yes	No
System Settings & Reset Demo Data	Yes	No	No

5. User Authentication & Session Security Requirements

1. Authentication Mode:

- Prototype: Local session persistence via `crm_session` with 1-click persona switchers for rapid user testing.

- Production Target: OAuth2 / JWT (JSON Web Tokens) with cryptographically signed bearer tokens, HTTP-only refresh cookies, and 15-minute token expiry.
2. **Session Timeout:** Automatic logout warning after 30 minutes of idle inactivity.
 3. **Route Guarding:** Client-side router checks `Auth.requireAuth([allowedRoles])` before rendering any page. Unauthorized navigation immediately redirects to `index.html`.
 4. **Audit Trail Association:** Every create, update, stage transition, and payment record must automatically stamp `userId`, `userName`, and `timestamp` from the active session.

Product Requirements Document (PRD)

`03_PRD_PRODUCT_REQUIREMENTS_DOCUMENT.md`

Product Requirements Document (PRD)

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** PRD-APEX-03
- **Version:** 2.0
- **Status:** Approved
- **Audience:** Product Managers, Software Engineers, UI/UX Designers, QA Automation Engineers

1. Product Vision & Core Design Principles

1.1 Product Vision

Career Apex CRM is a high-speed, candidate-centric sales and placement management platform engineered specifically for career training academies, college placement consultancies, and vocational staffing firms. It combines the density and agility of an Excel spreadsheet with the governance, auditability, and financial tracking of an enterprise CRM.

1.2 Core Product Principles

1. **Spreadsheet Velocity Over Complex UI:** Counselors spend 8 hours a day making 50+ outreach touches. The system prioritizes dense, Excel-style spreadsheet tables with instant 1-click actions over multi-layer navigation.
2. **Zero Corporate Account Clutter:** Student job seekers are individual candidates; the CRM strips away all corporate B2B "Account" fields, focusing solely on qualifications, passing years, target roles, and placement fees.
3. **Financial Transparency & Discipline:** Every candidate's contracted placement fee is mapped to milestone schedules, transactions, and printable voucher receipts with automated balance recalculation.
4. **Mandatory Operational Auditability:** Stage changes, counselor reassignments, and payment modifications must capture reasons, timestamps, and author identities.

2. Detailed Functional Requirements (FR)

FR-01: Candidate Profile & Educational Record Management

- **Description:** Centralized repository for all registered student job seekers.
- **Data Attributes:**
 - Candidate ID: Auto-generated unique string (CRM-STU-XXX).
 - Personal Information: Full Name, Primary Mobile (10 digits), Alternate Mobile/WhatsApp, Email, Permanent Address, City, State, Country.
 - Academic Background: Qualification/Degree, College/University, Passing Year (YOP: 2022–2026), CGPA or Percentage.
 - Career & Job Preferences: Target Job Role, Key Technical Skills, Experience Level (Fresher , 0-1 yr , 1-2 yrs , 2+ yrs), Expected CTC, Preferred Work Location, Resume Link.
 - Pipeline & Assignment: Pipeline Stage, Placement Status, Priority (High , Medium , Low), Lead Source, Assigned Sales Manager, Assigned Career Counselor.
 - Counseling Notes: Free-text counselor observations and interview readiness notes.
- **Validation:** Primary mobile must be exactly 10 numeric digits. Duplicate checks verify both primary and alternate mobile numbers against existing CRM records.

FR-02: Placement Fee & Payment Plan Generation

- **Description:** Commercial monetization engine managing agreed placement fees and scheduled installment milestones.
- **Fee Assignment:** Custom agreed total fee per student (default ₹45,000, customizable per candidate).
- **Payment Plans:**
 - Full Payment Upfront : 1 Milestone (100% due upon enrollment).
 - 2 Installments : Milestone 1 (50% upfront), Milestone 2 (50% due in 30 days or on placement).
 - 3 Installments : Milestone 1 (40% upfront), Milestone 2 (30% mid-training), Milestone 3 (30% on placement).
 - 4 Installments : Milestone 1 (25%), Milestone 2 (25%), Milestone 3 (25%), Milestone 4 (25%).
- **Milestone Properties:** Installment Number, Label, Due Date, Due Amount, Paid Date, Status (Pending , Paid , Overdue), Transaction Reference.

FR-03: Payment Transaction Ledger & Receipt Issuance

- **Description:** Financial recording of fees received and instant receipt generation.
- **Payment Attributes:** Unique Voucher Number (REC-XXXXX), Student ID, Student Name, Amount Paid (₹), Payment Mode (UPI , Net Banking , Credit/Debit Card , Cash , Cheque), Transaction / UTR Reference Number, Received by Counselor, Notes, Payment Timestamp.
- **Automatic Recalculation:**
 - $\text{\$paidAmount} = \sum \text{\$Payment Transactions}$

- $\text{\text{\text{pendingAmount}}} = \max(0, \text{\text{\text{totalFee}}} - \text{\text{\text{paidAmount}}})$
- Payment Status Transition:
 - If $\text{\text{\text{paidAmount}}} == \text{\text{\text{totalFee}}}$: Status = Fully Paid .
 - If $\text{\text{\text{paidAmount}}} > 0$ and $\text{\text{\text{paidAmount}}} < \text{\text{\text{totalFee}}}$: Status = Partially Paid .
 - If $\text{\text{\text{paidAmount}}} == 0$: Status = Pending .
 - If any unpaid installment due date $< \text{\text{\text{currentDate}}}$: Flag = Overdue .
- **Official Receipt Generator:** Formatted voucher modal with print trigger (`window.print()`), displaying candidate credentials, receipt voucher number, amount in words, counselor signature line, and official seal.

FR-04: Multi-Stage Pipeline & Excel Sheet Interface

- **Description:** Tabular spreadsheet workspace organizing candidates into distinct placement pipeline stages.
- **Pipeline Stages (10 Stages):**
 1. Cold Calling (Raw Data) : Uncalled database imported from college drives.
 2. New Lead : Fresh inquiries from web, walk-ins, or social media.
 3. Contacted : Initial outreach completed; counseling in progress.
 4. Interested : Candidate expressed interest in career placement track.
 5. Prospect : Training track demonstrated; placement fee discussed.
 6. Follow-up : Scheduled call or counseling session pending.
 7. Negotiation : Finalizing agreed fee amount and installment plan.
 8. Converted (Placed) : Placed in corporate role; successful conversion.
 9. Not Interested : Candidate declined services.
 10. Lost : Unresponsive, dropped out, or disqualified.
- **Interface Structure:**
 - Sidebar Submenu: Collapsible accordion linking directly to filtered stage views.
 - Stage Pill Ribbon: Horizontal pill buttons with dynamic count badges for quick sheet switching.
 - Sticky Actions Column: Quick icons for Call, WhatsApp, Email, View Dossier, and Edit.

FR-05: Stage Transition Audit & Mandatory Reason Capture

- **Description:** Prevents unverified stage movements by requiring an explicit business reason.
- **Functional Flow:**
 1. User triggers stage change via dropdown or inline table action.
 2. Stage Transition Modal opens displaying Student Name, Current Stage, and New Stage Selector.
 3. Transition Reason field is **mandatory**.
 4. Upon saving, system updates candidate stage, logs to `crm_stage_history` , and creates an entry in `crm_activities` .

FR-06: Bulk Ingestion Engine (Excel / CSV) with Deduplication

- **Description:** Mass import of raw candidate databases.
 - **Supported Formats:** `.xlsx`, `.xls`, `.csv` processed client-side via SheetJS.
 - **3-Step Wizard:**
 - *Step 1: Upload & Configuration:* Select target pipeline stage (e.g. `Cold Calling`) and target counselor allocation (direct or auto round-robin).
 - *Step 2: Intelligent Header Mapping:* Auto-matches uploaded sheet headers to CRM attributes (Name, Mobile, Email, College, Degree, YOP, Skills, Role, Fee).
 - *Step 3: Deduplication & Validation:* Validates 10-digit mobile syntax; flags existing CRM phone numbers as `Duplicate Phone (Will Skip)`.
 - **Commit:** Imports valid rows, assigns counselor, generates default fee plan, logs audit event, and redirects to pipeline sheet.
-

FR-07: Telephony Logging & Call Outcome Capture

- **Description:** Telephony disposition logging for compliance and productivity tracking.
 - **Functional Capabilities:**
 - Click-to-call initiates `tel:+91XXXXXXXXXX` protocol.
 - Call Outcome Modal captures: Duration (seconds/minutes), Disposition (`Connected - Interested`, `Ringing - No Answer`, `Busy`, `Invalid Number`, `Follow-up Requested`), and Discussion Notes.
 - Automatically updates `lastContacted` timestamp and appends to student call ledger.
-

FR-08: WhatsApp Direct Connect & Message Templating

- **Description:** Frictionless messaging via official WhatsApp deep-linking.
 - **Capabilities:**
 - Generates deep-link `https://wa.me/91XXXXXXXXXX?text=...`.
 - Template Library: Welcome Introduction, Course & Placement Pitch, Interview Schedule Confirmation, Fee Installment Reminder, Document Submission Alert.
 - Logs WhatsApp dispatch to student activity timeline.
-

FR-09: Multi-Channel Communication Center

- **Description:** Dedicated hub managing communication touchpoints.
 - **Features:** Call history ledger, WhatsApp dispatch logs, and email composer drawer with pre-configured counselor signatures.
-

FR-10: Follow-up & Task Agenda Scheduler

- **Description:** Calendar and task reminder engine.

- **Features:**

- Schedule follow-up with Date, Time, Priority (High , Medium , Low), Action Type (Call , WhatsApp , Email , Counseling), and Purpose.
- Categorization into Overdue (red badge), Today (amber badge), and Upcoming (slate badge).
- 1-click completion toggle with outcome notes capture.

FR-11: Candidate Reassignment Engine

- **Description:** Bulk or individual candidate transfer between counselors.
- **Capabilities:**
 - Admin can reassign any candidate to any manager or counselor.
 - Sales Manager can reassign candidates among reporting team counselors.
 - Updates managerId , managerName , salespersonId , salespersonName , and logs reassignment audit trail.

FR-12: Universal Multi-Filter Intelligence & Analytics

- **Description:** Cross-cutting report engine with multi-dimensional filtering.
- **Filter Parameters:** Date Preset (Today , Yesterday , This Week , This Month , Last 30 Days , This Quarter , Custom Range), Counselor, Manager, Pipeline Stage, Fee Status (Fully Paid , Partially Paid , Pending , Overdue), and Lead Source.
- **Reporting Tabs:**
 - *Tab 1: Placement Fees & Revenue:* Booked Fee, Cash Collected, Pending Dues, Collection Rate %, Counselor Realization Leaderboard, Payment Mode Split Doughnut, Revenue by Stage Bar, Overdue Milestones Alert Table with direct Collect action.
 - *Tab 2: Counselor Performance:* Leads vs Converted comparison chart, call counts, follow-up execution rate, win rates.
 - *Tab 3: Stage Velocity:* Transition volume breakdown (From Stage $\rightarrow$ To Stage).
 - *Tab 4: Lead Sources:* Inflow share doughnut chart and channel conversion efficiency.

FR-13: Activity Logging & System Compliance Stream

- **Description:** Immutable system audit trail.
- **Tracked Events:** Student Registration, Profile Edit, Stage Transition, Fee Payment, Follow-up Scheduled/Completed, Call Logged, Lead Reassigned, Student Deleted.
- **Searchable Attributes:** Search by user, customer ID, action keyword, date range.

FR-14: Team Roster & Productivity Leaderboards

- **Description:** Hierarchical counselor management and target comparison.
- **Features:** Team roster cards, reporting lines, counselor quotas, and monthly conversion leaderboard.

FR-15: System Configuration & Demo Data Seeding

- **Description:** System parameter configuration and cache control.
- **Features:** Placement fee standard presets, localStorage byte usage counters, and 1-click demo data reset button re-seeding 20+ realistic student files.

3. Non-Functional Requirements (NFR)

ID	Requirement Category	Target Specification
NFR-01	Performance	Table rendering for 1,000 candidate records shall execute in $< 150\text{ ms}$. Search filter debounce $< 250\text{ ms}$.
NFR-02	Scalability	Architecture must support migration to PostgreSQL capable of handling 500,000+ candidates and 5,000 concurrent counselors.
NFR-03	Data Integrity	Primary mobile numbers must remain unique. Financial balance invariant ($\text{totalFee} = \text{paid} + \text{pending}$) must strictly hold.
NFR-04	Usability	Dense, spreadsheet-style views must allow inline actions within a single click without full-page navigation.
NFR-05	Security	Role-based route guards and data scoping at the model layer. No unauthorized counselor access to peer student records.
NFR-06	Auditability	All financial transactions and stage changes must store user ID, timestamp, voucher numbers, and transition reasons.
NFR-07	Browser & Mobile Responsiveness	Fully responsive and mobile-friendly across smartphones (320px–480px), tablets (768px–992px), and desktops (1200px+). Includes off-canvas slide drawer navigation with darkened backdrop, swipeable tables with touch inertia, 44px minimum touch targets, and iOS auto-zoom prevention.
NFR-08	Zero-Dependency Runtime	Prototype must run standalone on any web server or file system without external database or node server dependencies.

4. Edge Cases & Exception Handling

1. **Duplicate Phone on Import:** When an uploaded CSV row has a phone number matching an existing candidate or another row in the same file, the system marks the row as duplicate, skips ingestion, and logs the skipped record in the import summary.
2. **Overpayment Attempt:** When a user attempts to record a payment amount exceeding the pending balance, the modal displays an error: *"Payment amount cannot exceed the pending due balance of ₹X."*
3. **Invalid Due Dates:** Milestone due dates cannot be set to a date preceding the student registration date.

4. **Session Expiry / Direct Access:** If an unauthenticated user opens an internal page (`pages/customers.html`), the route guard intercepts the request and redirects to `index.html` .
5. **Deletion of Placed Student:** Super administrator receives a high-severity confirmation prompt warning that deleting a placed student permanently removes associated ledger vouchers.
6. **Mobile Viewport Overflow:** Large Excel spreadsheet views and data tables must maintain touch scrolling without displacing the application header, drawer, or modal structures.

5. Acceptance Criteria & Definition of Done (DoD)

- [x] All 10 pipeline stages are selectable via sidebar submenus and top pill tabs.
- [x] Student registration enforces 10-digit mobile number uniqueness.
- [x] Bulk upload parses `.xlsx` and `.csv` files and assigns leads to counselors.
- [x] Agreed placement fees generate upfront and installment milestone schedules.
- [x] Payment recording updates ledger, recalculates balances, and produces printable receipts.
- [x] Reports provide universal multi-dimensional filters that dynamically refresh charts and tables.
- [x] Role-based scoping correctly filters data for Admin, Manager, and Counselor.
- [x] Mobile & Responsive Experience fully certified: slide-in drawer, touch backdrop, swipeable tables, and adaptive filter grids across iPhone, Android, and iPad viewports.

Screen-by-Screen Functional Specification

04_SCREEN_BY_SCREEN_FUNCTIONAL_SPECIFICATION.md

Screen-by-Screen Functional Specification

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** SCREEN-APEX-04
- **Version:** 2.0
- **Status:** Approved
- **Audience:** Frontend Developers, QA Engineers, UI/UX Designers, Product Owners

Screen Directory Overview

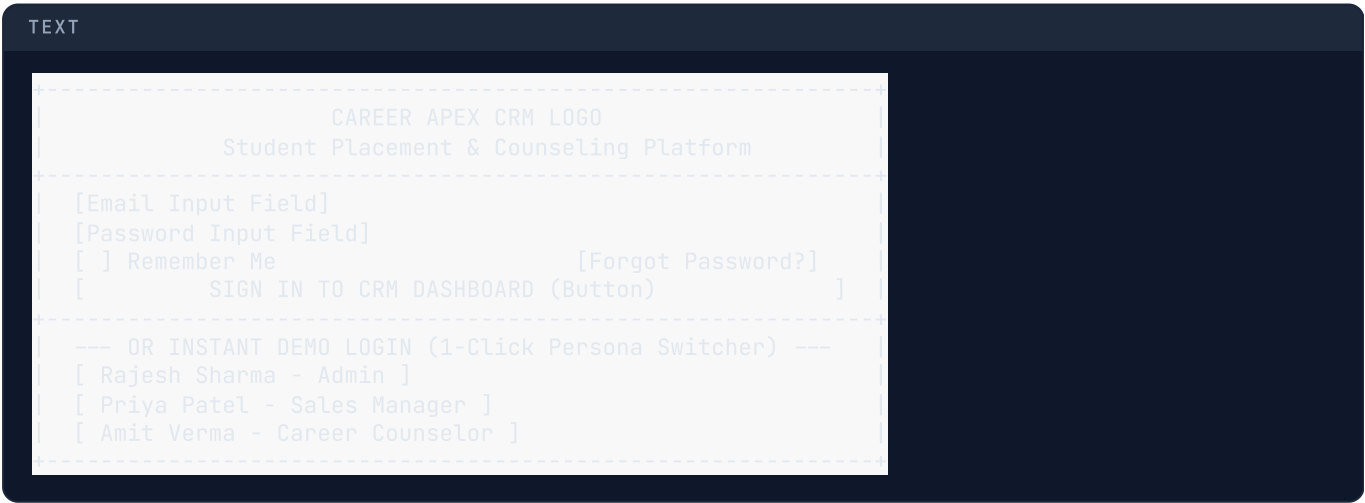
#	Screen Name	File Path	Route / URL	Target Roles
01	Login & Role Selector	index.html	/index.html	Public / All
02	Executive Admin Dashboard	pages/admin-dashboard.html	/pages/admin-dashboard.html	Admin
03	Sales Manager Command Center	pages/manager-dashboard.html	/pages/manager-dashboard.html	Manager, Admin
04	Counselor Daily Workspace	pages/salesperson-dashboard.html	/pages/salesperson-dashboard.html	Counselor (Sales), Manager, Admin
05	Student Job Seekers Directory	pages/customers.html	/pages/customers.html	All (Scoped)
06	Student Profile Dossier	pages/customer-details.html	/pages/customer-details.html?id=CRM-STU-XXX	All (Scoped)
07	Excel Spreadsheet Pipeline	pages/pipeline.html	/pages/pipeline.html?stage=...	All (Scoped)
08	Bulk Lead Import Engine	pages/import-leads.html	/pages/import-leads.html	Admin, Manager
09	Follow-up & Task Agenda	pages/followups.html	/pages/followups.html	All (Scoped)
10	Communication Center	pages/communication.html	/pages/communication.html	All (Scoped)
11	Audit Trail & Activity Log	pages/activities.html	/pages/activities.html	All
12	Sales Team & Counselor Roster	pages/team.html	/pages/team.html	Admin, Manager
13	Placement Intelligence Reports	pages/reports.html	/pages/reports.html	Admin, Manager
14	Settings & Demo Configuration	pages/settings.html	/pages/settings.html	Admin

Screen 01: Login & Role Selector (index.html)

1.1 Screen Purpose & Objective

Serves as the primary security gate for user authentication, and provides a 1-click demo persona switcher for testing different organizational perspectives (Admin, Manager, Counselor) without manually typing credentials.

1.2 Layout & Component Breakdown



1.3 UI Controls & Inputs

- **Email Field** (login-email): Required text input, validated for email syntax.
- **Password Field** (login-password): Password masked input.
- **Demo Switcher Buttons:**
 - btn-demo-admin : Logs in Rajesh Sharma (USR-001), redirects to admin-dashboard.html .
 - btn-demo-manager : Logs in Priya Patel (USR-002), redirects to manager-dashboard.html .
 - btn-demo-sales : Logs in Amit Verma (USR-004), redirects to salesperson-dashboard.html .

1.4 State Mutations & Data Flow

- Sets active session into crm_session with user object { id, name, email, role, avatar } .
- Sets crm_session_timestamp for timeout tracking.
- Redirects user based on user.role .

Screen 02: Executive Admin Dashboard (pages/admin-dashboard.html)

2.1 Screen Purpose & Objective

Provides executive leadership with real-time visibility into overall candidate intake, placement revenue, cash collected, pipeline velocity, and counselor performance across all centers.

2.2 Layout & Component Breakdown

1. **Header Strip:** Welcome greeting, quick action buttons: "Detailed Reports" $\rightarrow$ `reports.html`, "Open Pipeline" $\rightarrow$ `pipeline.html`.
2. **Placement Fee & Revenue Realization Banner:**
 - Total Booked Fee (`dash-fee-booked`): Aggregate contracted fees.
 - Realized Cash Collected (`dash-fee-collected`): Cash received in accounts.
 - Pending Dues (`dash-fee-pending`): Outstanding balance + Overdue milestones badge.
 - Collection Realization Rate % with visual progress bar (`dash-fee-rate-bar`).
3. **Candidate Operational KPIs (8 Metric Cards):**
 - Total Students (`kpi-total-leads`)
 - New Leads (`kpi-new-leads`)
 - Contacted (`kpi-contacted`)
 - Interested (`kpi-interested`)
 - Prospects (`kpi-prospects`)
 - Converted Deals (`kpi-converted`) + Win Rate %
 - Lost / Dropped Candidates (`kpi-lost`)
 - Today's Follow-ups (`kpi-today-followups`) + Overdue count
4. **Analytics Charts Grid:**
 - Left: Pipeline Stage Breakdown (Chart.js Bar Chart).
 - Right: Pipeline Stage Share (Chart.js Doughnut Chart).
5. **Team Leaderboard & Live Audit Trail:**
 - Left: Counselor Performance Leaderboard table (Counselor, Manager, Leads, Calls, Converted, Win Rate %).
 - Right: Real-time System Audit Feed (`recent-activities-list`).

Screen 03: Sales Manager Command Center (`pages/manager-dashboard.html`)

3.1 Screen Purpose & Objective

Enables regional and center sales managers to oversee their direct reporting counselors, monitor team revenue realization, reassign leads, and track daily scheduled follow-ups.

3.2 Layout & Component Breakdown

1. **Header Strip:** Greeting with dynamic `Manager: [Name]` badge, "Reassign Leads" and "Team Pipeline" buttons.
2. **Team Placement Revenue Banner:**
 - Team Booked Fee (`mgr-fee-booked`)
 - Team Cash Collected (`mgr-fee-collected`)
 - Team Pending Dues (`mgr-fee-pending`) + Overdue Milestones badge

- Team Collection Rate % with progress bar (mgr-fee-rate-bar)

3. **Team Pipeline KPIs:** Direct Team Members, Total Team Students, Today's Follow-ups, Interested Leads, Active Prospects, Team Converted.

4. Visuals & Team Reps Grid:

- Team Pipeline Distribution Chart.
- Reporting Counselors Table (mgr-team-reps-tbody) with assigned counts, converted counts, and a direct "View Leads" action button.

5. Actionable Feeds:

- Urgent Team Follow-ups queue (mgr-today-followups-list).
- Team Activity Audit Feed (mgr-team-activities-list).

Screen 04: Career Counselor Daily Workspace (pages/salesperson-dashboard.html)

4.1 Screen Purpose & Objective

The daily operational cockpit for individual counselors to execute cold calls, follow-up calls, WhatsApp messaging, and record student installment payments.

4.2 Layout & Component Breakdown

1. **Header Strip:** Counselor greeting with role badge, "My Pipeline" $\rightarrow$ pipeline.html, "My Follow-ups" $\rightarrow$ followups.html.

2. Personal Target & Collection Banner:

- Personal Booked Fees (rep-fee-booked)
- Personal Cash Collected (rep-fee-collected)
- Personal Pending Dues (rep-fee-pending) + Overdue Milestones badge
- Personal Realization Rate % (rep-fee-rate) with progress bar

3. **Personal Pipeline KPIs:** My Leads, Today's Follow-ups, Overdue Follow-ups, Interested, Prospects, Closed Placements.

4. Today's Action Tasks Queue (rep-today-tasks-container):

- High-priority task cards grouped by urgency (Overdue tasks highlighted in red).
- 1-Click Action Buttons on each card:
 - Call : Opens telephony dialer.
 - WhatsApp : Opens WhatsApp Web with pre-filled message.
 - Email : Opens email drawer.
 - Complete : Marks task done with outcome notes.

5. **My Active Portfolio Table (rep-recent-leads-tbody):** Quick-reference list of assigned candidates and current stages.

Screen 05: Student Job Seekers Directory (pages/customers.html)

5.1 Screen Purpose & Objective

Master searchable and filterable candidate directory providing comprehensive access to student profiles, qualifications, stages, placement fees, and counselor assignments.

5.2 Multi-Filter Ribbon Controls

- **Row 1:**
 - Search Keyword: Live search matching Student ID, Name, College, Qualification, Skills, Mobile.
 - Pipeline Stage: Filter by any of the 10 stages or **All Stages**.
 - Placement Status: **All**, **Active**, **In Discussion**, **Placed / Hired**, **Dropped / Inactive**.
 - Priority: **All**, **High**, **Medium**, **Low**.
- **Row 2:**
 - Career Counselor: Dropdown of all sales reps.
 - Passing Year (YOP): **2026**, **2025**, **2024**, **2023**, **2022**.
 - City: Dynamic list of candidate cities.
 - **Placement Fee Status:** **All**, **Fully Paid**, **Partially Paid**, **Pending**, **Overdue**.
 - Rows per page: **10**, **25**, **50**.
 - Clear Filters button: Resets all inputs and reloads table.

5.3 Directory Table Columns

1. **Student ID** : Clickable link $\rightarrow$ **customer-details.html?id=...**
2. **Candidate & Degree** : Student Name + Qualification/Degree.
3. **College / University** : College name + Passing Year (YOP) + City.
4. **Target Role & Skills** : Preferred job title + skills tags.
5. **Mobile Phone** : Formatted 10-digit number.
6. **Stage** : Stage badge (e.g. **Cold Calling**, **Interested**).
7. **Placement Fee & Dues** : Bold total fee, paid vs pending balance, and payment status badge (**Fully Paid**, **Partially Paid**, **Pending**, **Overdue**).
8. **Status** : Status badge (**Active**, **Converted**, **Lost**).
9. **Priority** : Priority badge.
10. **Counselor** : Assigned counselor name + reporting manager.
11. **Next Follow-up** : Date formatted (colored red if past current date).
12. **Quick Actions** : View Profile (Eye), Edit Profile (Pencil), Call (Phone), WhatsApp (Green icon), Email (Envelope), Delete (Trash - Admin only).

5.4 Modals

- **Edit Student Profile Modal (**edit-customer-modal**):**
 - Section 1: Candidate Details (Name, Email, Mobile, Alternate Mobile).
 - Section 2: Academic Background (Degree, College, YOP, Experience Level, CGPA/%).

- Section 3: Job Preferences & Skills (Target Role, Skills, City, State, Location, CTC, Address).
- Section 4: Pipeline & Assignment (Stage, Status, Priority, Source, Manager, Counselor).
- **Section 5: Placement Fee & Payment Plan:** Agreed Placement Fee (₹), Payment Plan (Full Upfront , 2 Installments , 3 Installments , 4 Installments), Payment Status (Pending , Partially Paid , Fully Paid).
- Section 6: Counseling Notes.

Screen 06: Student Profile Dossier (pages/customer-details.html)

6.1 Screen Purpose & Objective

The central, complete record for an individual student candidate. Aggregates identity, academic history, counseling notes, call logs, follow-ups, stage transitions, and placement fee financial ledgers.

6.2 Hero Identity Banner

- Candidate Avatar with initials.
- Full Name, Student Code (CRM-STU-XXX), Primary Phone, WhatsApp Icon, Email, City, Expected CTC.
- **Stage Progression Dropdown:** Allows moving candidate to any of the 10 stages; prompts mandatory reason modal.
- **Hero Action Toolbar:** Call, WhatsApp, Email, Schedule Follow-up, Edit Profile.

6.3 Seven Tabbed Dossier Sections

Tab 1: Overview & Academic Profile (tab-overview)

- Academic Background Card: College, Degree, Passing Year, CGPA, Experience.
- Job Preferences Card: Target Role, Technical Skills, Preferred Location, Expected CTC.
- **Placement Fee Summary Card:** Total Agreed Fee, Cash Paid, Outstanding Balance, Payment Plan, Status badge, and quick link to "View Full Ledger".
- Recent Counseling Notes preview.

Tab 2: Activity Timeline (tab-timeline)

- Chronological list of all system interactions with timestamps, action badges, and author names.

Tab 3: Counseling Notes (tab-notes)

- Rich text notes input area + historical notes stream showing counselor remarks on candidate attitude, interview feedback, and skill readiness.

Tab 4: Call Logs (tab-calls)

- Log Phone Call button launching call outcome modal.
- Call history table: Timestamp, Counselor, Duration, Call Disposition, Outcome Notes.

Tab 5: Scheduled Follow-ups (tab-followups)

- Schedule Follow-up button.

- Follow-ups table with status badges (Pending , Completed), purpose, date, time, and 1-click completion toggle.

Tab 6: Stage History (tab-stage-history)

- Audit log tracking every pipeline progression: Old Stage $\rightarrow$ New Stage, Transition Reason, Counselor Name, Timestamp.

Tab 7: Placement Fee & Payments (tab-payments)

- 4 Financial KPI Cards:**
 - Total Agreed Placement Fee (fee-kpi-total) in ₹.
 - Total Collected to Date (fee-kpi-paid) in ₹.
 - Outstanding Due Balance (fee-kpi-pending) in ₹.
 - Payment Status Badge & Plan (fee-kpi-status).
- Collection Progress Bar:** Visual progress representation of % fee realized.
- Scheduled Installment Milestones Table:** Milestone #, Milestone Label, Due Date (flagged red if overdue), Amount, Milestone Status (Pending , Paid , Overdue), and a direct "Collect Payment" action button.
- Payment Transaction Ledger Table:** Transaction Code (REC-XXXXX), Date, Amount Paid (₹), Payment Mode (UPI , Net Banking , Card , Cash , Cheque), Transaction / UTR Reference Number, Received by Counselor, and "View Receipt" button.

6.4 Modals Launched from Screen 06

- Record Placement Fee Payment Modal (modal-record-payment):**
 - Milestone Selector (Choose milestone or custom payment).
 - Payment Amount (pre-filled, validates against pending balance).
 - Payment Mode: UPI , Net Banking , Credit/Debit Card , Cash , Cheque .
 - Transaction / UTR Reference Number input.
 - Counselor receiver selector.
 - Payment Notes.
- Official Payment Receipt Modal (modal-view-receipt):**
 - Formatted printable voucher containing: Career Apex CRM Header, Voucher Number, Issue Date, Student Name, Student ID, Mobile, Course/Placement Track, Amount Paid in Numbers and Words, Payment Mode, UTR Reference Number, Remaining Balance, Counselor Signature Line, Official Stamp, and "Print Receipt" button triggering window.print() .
- Log Call Outcome Modal (modal-log-call).**
- Schedule Follow-up Modal (modal-add-followup).**
- Stage Transition Modal (modal-change-stage).**

Screen 07: Excel Spreadsheet Pipeline View (pages/pipeline.html)

7.1 Screen Purpose & Objective

Replaces cumbersome Kanban boards with a dense, horizontal Excel-style spreadsheet. Optimized for counselors to work down a list of 100+ candidates with instant stage moves and communications.

7.2 Component Breakdown

- 1. Top Stage Pill Ribbon:** Horizontal scrolling pill bar displaying `All Leads` and all 10 stages (`Cold Calling` , `New Lead` , `Contacted` , `Interested` , `Prospect` , `Follow-up` , `Negotiation` , `Converted` , `Not Interested` , `Lost`) with live candidate counts. Clicking a pill filters the sheet instantly and updates the browser URL (`?stage=Cold Calling`).
- 2. Excel Ribbon Toolbar:**
 - Quick Search: Searches candidate name, college, degree, skills, role, phone.
 - Passing Year Filter: `2026` , `2025` , `2024` , `2023` , `2022` .
 - Priority Filter: `High` , `Medium` , `Low` .
 - Counselor Filter: Dropdown of sales reps.
 - **Fee Status Filter:** `All` , `Fully Paid` , `Partially Paid` , `Pending` , `Overdue` .
 - Reset Filters button.
- 3. Excel Spreadsheet Table (`excel-pipeline-table`):**
 - Columns: `#` (Row Index), Student ID, Student Name, Degree, College, YOP, Key Skills, Target Role, Experience, Mobile, Quick Connect (Call/WhatsApp/Email), Pipeline Stage (Clickable to change), Priority, Status, **Placement Fee & Due Balance**, Career Counselor, City, Next Follow-up.
 - Sticky Actions Column: Profile, Edit, Quick Reassign, Quick Note.
- 4. Excel Status Bar:** Shows system status (`READY`), visible row count, active stage name, total enrolled candidate count, and overall placement conversion win rate %.

Screen 08: Bulk Lead Import Engine (pages/import-leads.html)

8.1 Screen Purpose & Objective

Mass ingestion of raw student records from university placement cells or job portal downloads.

8.2 Three-Step Wizard Architecture

- **Step 1: File Upload & Target Configuration:**
 - Drag-and-drop zone supporting `.xlsx` , `.xls` , and `.csv` .
 - **Target Pipeline Stage Selector:** Allows directing imported candidates specifically into `Cold Calling (Raw Data)` or `New Lead` .
 - **Target Counselor Allocation:** Choose "Auto Round-Robin Distribution" (spreads evenly across team) or select a specific counselor.
- **Step 2: Intelligent Header Mapping:**

- Maps uploaded spreadsheet column headers to CRM candidate attributes: Student Name, Mobile, Alternate Mobile, Email, Degree/Qualification, College, Passing Year, Skills, Target Role, City, State, Placement Fee.
- **Step 3: Deduplication, Validation & Preview:**
 - Parses rows client-side via SheetJS.
 - Validates 10-digit mobile syntax.
 - Checks phone numbers against existing CRM records.
 - Displays validation status tags: `Valid (Ready to Import)`, `Duplicate Phone (Will Skip)`, `Invalid Mobile`.
- **Commit Import Button:** Writes valid candidate objects, sets up initial placement fee plans, records system audit event, and redirects to the pipeline view.

Screen 09: Follow-up & Task Agenda (`pages/followups.html`)

9.1 Screen Purpose & Objective

Daily task execution hub organizing scheduled callbacks, counseling interviews, and fee reminder tasks.

9.2 Component Breakdown

- Top KPI Summary: Today's Tasks, Overdue Tasks, Upcoming Tasks, Completed to Date.
- Filter Bar: Date filters, Priority filters, Counselor filters.
- Task Lists categorized by urgency:
 - **Overdue Follow-ups (Red Border):** Tasks whose scheduled date/time is in the past.
 - **Today's Follow-ups (Amber Border):** Tasks scheduled for the current date.
 - **Upcoming Follow-ups (Slate Border):** Tasks scheduled for future dates.
- Interactive Controls: 1-click Call, WhatsApp, Email, and Complete checkbox.

Screen 10: Communication Center (`pages/communication.html`)

10.1 Screen Purpose & Objective

Centralized multi-channel outreach workspace for phone calls, WhatsApp campaigns, and email dispatches.

10.2 Component Breakdown

- Telephony dialer widget with recent calls list.
- WhatsApp message builder with pre-configured counseling templates.
- Email dispatch history and template selector.

Screen 11: Real-time Activity & Audit Trail ([pages/activities.html](#))

11.1 Screen Purpose & Objective

Comprehensive, searchable audit log ensuring regulatory compliance and tracking all user actions.

11.2 Component Breakdown

- Search and filter bar (Filter by Counselor, Action Type, Student ID, Date).
 - Chronological timeline feed rendering: User avatar, Action badge, Student name, Timestamp, and Detailed description.
-

Screen 12: Sales Team & Counselor Hierarchy ([pages/team.html](#))

12.1 Screen Purpose & Objective

Roster management, manager reporting lines, and counselor performance scorecards.

12.2 Component Breakdown

- Add Team Member modal (Name, Email, Role, Reporting Manager).
 - Counselor profile cards showing avatar, reporting manager, active candidate load, conversion count, and win rate %.
 - Monthly performance comparison leaderboard.
-

Screen 13: Sales Intelligence & Placement Reports ([pages/reports.html](#))

13.1 Screen Purpose & Objective

Multi-dimensional reporting suite providing analytics across financial fee realization, team benchmarks, stage transitions, and lead sources.

13.2 Universal Multi-Dimensional Filter Toolbar

- Date Preset: **All Time** , **Today** , **Yesterday** , **This Week** , **This Month** , **Last 30 Days** , **This Quarter** , **Custom Range** .
- Start Date & End Date pickers.
- Career Counselor dropdown.
- Sales Manager dropdown.
- Pipeline Stage dropdown.
- **Fee / Payment Status dropdown:** **All** , **Fully Paid** , **Partially Paid** , **Pending** , **Overdue** .
- Lead Source dropdown.
- Apply Filters & Reset Filters buttons.
- Export Active Report CSV button.

13.3 Four Reporting Tabs

- **Tab 1: Placement Fees & Revenue Analytics:**
 - 4 KPI Cards: Total Booked Fee, Cash Collected, Pending Dues, Collection Rate %.
 - Counselor Fee Realization Leaderboard Table (Counselor, Manager, Students, Booked, Collected, Balance, Realization %).
 - Payment Methods Doughnut Chart (UPI, Net Banking, Card, Cash).
 - Revenue Realization by Stage Bar Chart.
 - **Overdue & Pending Milestones Alert Table:** Displays candidates with upcoming or overdue installments, counselor name, due date, amount, and direct "Collect" action.
- **Tab 2: Counselor Performance:** Comparison Bar Chart (Assigned Leads vs Closed Placements) and detailed breakdown table.
- **Tab 3: Stage Movement & Velocity:** Transition breakdown table (From Stage $\rightarrow$ To Stage) and transition volume bar chart.
- **Tab 4: Lead Source Attribution:** Doughnut chart of lead inflow channels and conversion rate table.

Screen 14: System Settings & Demo Configuration ([pages/settings.html](#))

14.1 Screen Purpose & Objective

System administrative settings, storage quota management, and demo re-seeding controls.

14.2 Component Breakdown

- Organization profile and branding configuration.
- Placement fee defaults (Standard fee amount, default installment split).
- Storage statistics: Real-time calculation of browser localStorage bytes utilized and total object counts.
- **Reset Demo Data Action:** Clears cache and re-seeds 20+ realistic student candidate files, complete with academic credentials, payment ledgers, call logs, and activities.

High-Level Architectural Design (HLD)

05_HLD_HIGH_LEVEL_DESIGN.md

High-Level Design Document (HLD)

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** HLD-APEX-05
- **Version:** 2.0
- **Status:** Approved
- **Audience:** System Architects, Tech Leads, Engineering Managers, Security Officers

1. Architectural Vision & System Topology

1.1 Architectural Philosophy

Career Apex CRM is designed with a **Strict Layered Separation of Concerns**. The current frontend prototype operates with pure client-side persistence (`StorageService`), but its software interfaces (Data Access, Model Contracts, and Service Facades) are structured to mirror standard enterprise REST API boundaries. This guarantees that transitioning to a full-stack backend requires zero redesign of the visual UI components.

1.2 Enterprise Production Architecture (C4 Container Diagram)

```

graph TD
    User["User["Career Counselor / Manager / Admin  
(Web Browser)"]"]

    subgraph EdgeTier [Edge & Delivery Tier]
        CDN["CDN["Cloudflare CDN / Edge Workers  
(SSL Termination, DDoS, Asset Caching)"]"]
        WAF["WAF["Web Application Firewall (WAF)"]"]
    end

    subgraph IngressTier [Ingress & Gateway Tier]
        ReverseProxy["ReverseProxy["Nginx / Traefik Reverse Proxy  
(Load Balancer, Rate Limiting)"]"]
        APIGateway["APIGateway["API Gateway & Auth Verification"]"]
    end

    subgraph AppTier [Application Server Tier - NestJS / Spring Boot]
        AuthService["AuthService["Auth & RBAC Service"]"]
        StudentService["StudentService["Student Candidate Service"]"]
        PipelineService["PipelineService["Pipeline & Stage Engine"]"]
        PaymentService["PaymentService["Placement Fee & Ledger Service"]"]
        CommService["CommService["Communication Adapter (WhatsApp/Tel/Email)"]"]
        ReportService["ReportService["Reporting & Analytics Engine"]"]
        AuditService["AuditService["Activity & Audit Logger"]"]
    end

    subgraph CacheTier [In-Memory & Queue Tier]
        RedisCache["RedisCache["Redis 7 Cluster  
(Session Cache, Dashboard KPI Store)"]"]
        WorkerQueue["WorkerQueue["BullMQ Background Workers  
(Bulk CSV Ingestion, Receipt PDF Generation)"]"]
    end

    subgraph DataTier [Persistence & Storage Tier]
        PrimaryDB["PrimaryDB[(PostgreSQL 16 Primary DB  
ACID Transactions, Read Replicas)]"]
        ObjectStore["ObjectStore["Cloudflare R2 / AWS S3  
(Resumes, Vouchers, CSV Backups)"]"]
    end

    User --> CDN --> WAF --> ReverseProxy --> APIGateway
    APIGateway --> AuthService
    APIGateway --> StudentService
    APIGateway --> PipelineService
    APIGateway --> PaymentService
    APIGateway --> CommService
    APIGateway --> ReportService
    APIGateway --> AuditService

    AuthService --> RedisCache
    StudentService --> PrimaryDB
    PipelineService --> PrimaryDB
    PaymentService --> PrimaryDB
    ReportService --> RedisCache
    ReportService --> PrimaryDB
    AuditService --> PrimaryDB

    StudentService --> WorkerQueue
  
```

WorkerQueue → PrimaryDB
PaymentService → ObjectStore

2. Subsystem & Module Decomposition

```
graph TD
    UI[Frontend User Interface Layer  
HTML5, CSS3, Vanilla ES6+ / React SPA]
    subgraph CoreModules [Core CRM Modules]
        M1[Auth & Session Module  
js/auth.js]
        M2[Student Directory Module  
js/customers.js]
        M3[Placement Fee & Payment Engine  
js/payments.js]
        M4[Excel Pipeline & Stage Module  
js/pipeline.js]
        M5[Multi-Channel Communication  
js/communication.js]
        M6[Universal Reporting Engine  
js/reports.js]
        M7[Task & Follow-up Scheduler  
js/followups.js]
        M8[Activity & Audit Logger  
js/activities.js]
    end
    subgraph InfrastructureLayer [Infrastructure & Storage Layer]
        Storage[StorageService Layer  
js/storage.js]
        Seed[Seed Data Initializer  
js/seed-data.js]
        Validation[Validation Utility  
js/validation.js]
        Utils[Common Formatting & Helpers  
js/utlis.js]
    end
    UI --> CoreModules
    CoreModules --> InfrastructureLayer
```

2.1 Module Responsibility Matrix

Module Name	File Path	Core Responsibilities
Auth & Session	js/auth.js	User authentication, demo persona switching, session storage, and route authorization guards.
Student Directory	js/customers.js	CRUD operations for student candidate files, academic qualifications, deduplication checks, and data scoping.
Placement Payments	js/payments.js	Agreed fee plans, milestone schedule generation, transaction ledger logging, balance calculation, and receipt generation.
Pipeline & Stages	js/pipeline.js	10-stage state transitions, mandatory reason enforcement, and Excel spreadsheet view rendering.
Communication	js/communication.js	Telephony call logging, WhatsApp deep-link generation with templates, and email composer.
Reporting & Analytics	js/reports.js	Universal multi-dimensional dataset filtering, fee realization calculations, team win rate benchmarks.
Follow-up Tasks	js/followups.js	Task scheduling, categorization (Overdue , Today , Upcoming), and completion outcome capture.
Activity Audit	js/activities.js	Immutable chronological logging of all candidate interactions, stage transitions, and financial events.
Storage Service	js/storage.js	Data abstraction layer wrapping storage persistence with automated ID generation and JSON serialization.

3. End-to-End Architectural Data Flows

3.1 Bulk Ingestion & Duplicate Protection Flow

```
sequenceDiagram
    autonumber
    actor User as Sales Manager / Admin
    participant UI as Import Wizard (import-leads.html)
    participant Parser as SheetJS Client Parser
    participant Model as Customers Model (customers.js)
    participant Storage as StorageService (storage.js)
    participant Audit as Activities Logger (activities.js)

    User->>UI: Selects Target Pipeline Stage ('Cold Calling') & Target Counselor
    User->>UI: Uploads candidates.xlsx (1,000 Rows)
    UI->>Parser: Parse File Buffers to JSON Array
    Parser->>UI: Array of Row Objects
    UI->>UI: User Maps Spreadsheet Columns to CRM Attributes
    loop For each candidate row
        UI->>Model: Validate 10-Digit Mobile Syntax
        UI->>Model: Check Duplicate Phone in Storage
        alt Duplicate Mobile Found
            Model->>UI: Flag Row: 'Duplicate Phone (Will Skip)'
        else Valid Unique Mobile
            Model->>UI: Flag Row: 'Valid (Ready to Import)'
        end
    end
    end
    User->>UI: Clicks 'Commit Import'
    UI->>Model: Create Student Candidate Objects
    Model->>Storage: Append to 'crm_customers'
    Model->>Storage: Seed Default Fee Milestones in 'crm_customers'
    Model->>Audit: Log 'Bulk Import Completed' Event
    UI->>User: Success Notification & Redirect to Pipeline Sheet
```

3.2 Stage Progression with Mandatory Audit Reason

```
sequenceDiagram
    autonumber
    actor Counselor as Career Counselor
    participant View as Pipeline Excel Sheet
    participant Modal as Stage Transition Modal
    participant Engine as Pipeline Engine (pipeline.js)
    participant Hist as Stage History Ledger
    participant Audit as Activities Logger

    Counselor->>View: Clicks Candidate Stage Badge (e.g. 'Interested')
    View->>Modal: Opens Modal with Candidate Info & Stage Dropdown
    Counselor->>Modal: Selects New Stage ('Prospect')
    Counselor->>Modal: Types Mandatory Reason ('Completed technical screening')
    Counselor->>Modal: Submits Form
    Modal->>Engine: updateStage(customerId, 'Prospect', reason, currentUser)
    Engine->>Engine: Mutate Candidate Object: stage = 'Prospect'
    Engine->>Hist: Append Record { fromStage: 'Interested', toStage: 'Prospect', reason, counselor }
    Engine->>Audit: Log Activity 'Stage Transition: CRM-STU-XXX moved to Prospect'
    Engine->>View: Re-render Spreadsheet Row & Update Status Bar
```

3.3 Fee Collection, Balance Recalculation & Receipt Issuance

```
sequenceDiagram
    autonumber
    actor Counselor as Career Counselor
    participant Details as Student Profile (customer-details.html)
    participant PayModal as Record Payment Modal
    participant PayEngine as Payments Engine (payments.js)
    participant Storage as StorageService
    participant ReceiptModal as Printable Receipt Modal

    Counselor->>Details: Clicks 'Collect Payment' on Milestone 1 (₹22,500)
    Details->>PayModal: Open Modal with Pre-filled Amount & Milestone Label
    Counselor->>PayModal: Enters Payment Mode ('UPI'), UTR Ref ('UTR-98234120'), Notes
    Counselor->>PayModal: Submits Payment
    PayModal->>PayEngine: recordPayment({ customerId, amount, mode, ref, receivedBy })
    PayEngine->>PayEngine: Generate Voucher Number ('REC-00124')
    PayEngine->>Storage: Append to 'crm_payments'
    PayEngine->>PayEngine: Update Milestone Status to 'Paid'
    PayEngine->>PayEngine: Recalculate: paidAmount += 22500, pendingAmount -= 22500
    PayEngine->>PayEngine: Update paymentStatus ('Partially Paid' or 'Fully Paid')
    PayEngine->>Storage: Update 'crm_customers'
    PayEngine->>Storage: Log Audit Activity
    PayEngine->>Details: Render Updated Balances & Ledger Table
    PayEngine->>ReceiptModal: Render Formatted Voucher with Candidate & Payment Details
    ReceiptModal->>Counselor: Displays Official Printable Receipt
```

4. Security & Access Control Architecture

4.1 Route Guarding & Client-Side Interceptors

Every page template initializes with an authentication checkpoint:

JAVASCRIPT

```
const user = Auth.requireAuth(['admin', 'manager'])
if (!user) return // Immediate redirect to index.html
```

4.2 Model-Level Data Scoping

Data visibility is enforced within the model, not merely hidden in CSS:

- Counselors calling `Customers.getScopedCustomers()` receive an array filtered strictly by `c.salespersonId === user.id`.
- Managers receive an array filtered by `c.managerId === user.id`.
- Direct ID lookups via `Customers.getById(id)` verify that the retrieved candidate belongs to the active user's authorization boundary.

4.3 XSS Sanitization & Data Protection

- All user-generated strings (Candidate Name, College, Skills, Notes, Reasons) pass through `Utils.escapeHtml()` before DOM injection to eliminate Cross-Site Scripting (XSS) attack vectors.
- Telephony numbers pass through strict regex sanitation (`^\d{10}$`) preventing injection payloads in communication deep-links.

5. Scalability & Performance Strategy

1. **Virtualization Strategy for Large Sheets:** When migrating to 50,000+ candidate sheets, the table renderer utilizes virtual windowing (rendering only visible rows in the viewport) to maintain sub-100ms render speeds.
2. **Asynchronous Background Ingestion:** Production backend delegates bulk Excel parsing and duplicate scanning to Redis-backed worker queues (`BullMQ`), preventing API Gateway timeouts on 100,000-row uploads.
3. **Optimistic UI Updates:** Client-side prototype updates UI immediately upon recording calls, follow-ups, and payments, writing to storage asynchronously.

Low-Level Component Design (LLD)

`06_LLD_LOW_LEVEL_DESIGN.md`

Low-Level Design Document (LLD)

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** LLD-APEX-06
- **Version:** 2.0
- **Status:** Approved
- **Audience:** Full-Stack Engineers, Frontend Developers, QA Automation Engineers

1. Class & Module Architecture

The application is structured into modular JavaScript classes operating under a singleton pattern, exposing predictable programmatic contracts.

```
classDiagram
class StorageService {
+getData(key, defaultVal)
+saveData(key, data)
+generateId(prefix)
+clearData(key)
}

class Auth {
+login(email, password)
+logout()
+getCurrentUser()
+requireAuth(allowedRoles)
}

class Customers {
+getAll()
+getScopedCustomers(user)
+getById(id)
+create(data)
+update(id, data)
+delete(id)
+filter(params)
+paginate(items, page, pageSize)
+checkDuplicatePhone(mobile, altMobile, excludeId)
}

class Payments {
+getAll()
+getByCustomerId(customerId)
+recordPayment(data)
+generateDefaultInstallments(totalFee, plan)
+updateFeePlan(customerId, planData)
+getSummary(scopedCustomers)
+getInstallmentDues(scopedCustomers, filter)
+getCounselorRevenueReport()
+generateReceiptHtml(payment, customer)
+formatCurrency(amount)
}

class Pipeline {
+getStages()
+getLeadsByStage(stage, user)
+updateStage(customerId, newStage, reason, user)
+openStageModal(customerId, newStage, callback)
}

class Followups {
+getAll(scoped)
+getByCustomerId(customerId)
+create(data)
+complete(id, outcomeNotes)
+getCategorized(scoped)
}

class Communication {
+initiateCall(customerId)
+logCall(data)
+openWhatsApp(customerId, templateKey)
+initiateEmail(customerId)
}
```

```

class Reports {
  +filterCustomers(filters)
  +getRevenueReport(filters)
  +getTeamPerformanceReport(filters)
  +getStageMovementReport(filters)
  +getDashboardKPIs(user)
  +getLeadSourceStats(filters)
}

StorageService <.. Customers
StorageService <.. Payments
StorageService <.. Pipeline
StorageService <.. Followups
StorageService <.. Communication
StorageService <.. Reports
Customers <.. Payments
Customers <.. Pipeline
Customers <.. Reports

```

2. Detailed Method Signatures & Programmatic Contracts

2.1 Payments Module (js/payments.js)

Payments.recordPayment(data)

- Parameters:

TYPESCRIPT

```

interface RecordPaymentPayload {
  customerId: string;
  amount: number;
  paymentMode: 'IIP' | 'Net Banking' | 'Credit/Debit Card' | 'Cash' | 'Cheque';
  transactionRef: string;
  receivedBy?: string;
  installmentId?: string;
  notes?: string;
}

```

- Returns:

TYPESCRIPT

```

interface RecordPaymentResult {
  success: boolean;
  message?: string;
  payment?: PaymentTransactionObject;
  customer?: StudentCandidateObject;
}

```

- Operational Logic:

1. Validates that `amount > 0`.
2. Retrieves customer; verifies `amount ≤ customer.pendingAmount`.
3. Generates unique voucher code: `StorageService.generateId('REC')`.

4. Appends payment record to `crm_payments` .
5. Updates matching installment milestone status to `Paid` and records `paidDate` and `transactionRef` .
6. Increments `customer.paidAmount` and decrements `customer.pendingAmount` .
7. Updates `customer.paymentStatus` : sets `Fully Paid` if `pendingAmount === 0` , else `Partially Paid` .
8. Commits customer updates to `crm_customers` .
9. Logs audit event to `crm_activities` .

`Payments.generateDefaultInstallments(totalFee, plan)`

- **Parameters:** `totalFee: number` , `plan: string`
- **Returns:** `InstallmentMilestone[]`
- **Installment Generation Logic:**
 - `Full Payment Upfront` :
 - Milestone 1: 100% of `totalFee` , due in 7 days.
 - `2 Installments` :
 - Milestone 1 (Registration): 50% of `totalFee` , due in 7 days.
 - Milestone 2 (On Placement / Final): 50% of `totalFee` , due in 45 days.
 - `3 Installments` :
 - Milestone 1 (Registration): 40% of `totalFee` , due in 7 days.
 - Milestone 2 (Mid-Course Training): 30% of `totalFee` , due in 30 days.
 - Milestone 3 (Corporate Placement): 30% of `totalFee` , due in 60 days.
 - `4 Installments` :
 - Milestones 1 to 4: 25% each, due in 7, 30, 60, and 90 days respectively.

2.2 Customers Module (`js/customers.js`)

`Customers.checkDuplicatePhone(mobile, altMobile, excludeId)`

- **Parameters:** `mobile: string` , `altMobile?: string` , `excludeId?: string`
- **Returns:**

TYPESCRIPT

```
interface DuplicateCheckResult {
  isDuplicate: boolean;
  type?: 'primary' | 'alternate';
  customer?: StudentCandidateObject;
}
```

- **Operational Logic:** Iterates through all candidates in storage. If `candidate.id !== excludeId` , compares numeric mobile strings against both `candidate.mobile` and `candidate.altMobile` .

Customers.filter(params)

- **Parameters:**

TYPESCRIPT

```
interface FilterParams {  
  search?: string;  
  stage?: string;  
  status?: string;  
  priority?: string;  
  salespersonId?: string;  
  managerId?: string;  
  passingYear?: string;  
  city?: string;  
  paymentStatus?: 'All' | 'Fully Paid' | 'Partially Paid' | 'Pending' | 'Overdue';  
}
```

- **Returns:** StudentCandidateObject[]

- **Filtering Logic:**

- Evaluates search keywords against ID, Name, College, Qualification, Skills, Role, Mobile, and City.
- Applies strict filters for Stage, Status, Priority, Counselor, Manager, YOP, City.
- **Special Overdue Evaluation:** When `paymentStatus === 'Overdue'`, checks whether any installment milestone has `(status === 'Pending' || status === 'Overdue')` and `dueDate < todayStr`.

3. Finite State Machine Specifications

3.1 Candidate Placement Pipeline State Machine

stateDiagram-v2

[*] → ColdCalling : Bulk Upload / Raw Sourcing

[*] → NewLead : Web Inflow / Walk-in Registration

ColdCalling → Contacted : Phone Call / WhatsApp Connect

ColdCalling → NotInterested : Candidate Declines Placement

ColdCalling → Lost : Invalid Number / Disconnected

NewLead → Contacted : Initial Qualification Outreach

NewLead → NotInterested : Declined

Contacted → Interested : Expressed Explicit Career Track Interest

Contacted → FollowUp : Scheduled Callback Requested

Contacted → NotInterested : Not Interested

Interested → Prospect : Training Track Demo Given & Placement Fee Agreed

Interested → FollowUp : Needs Family / College Clearance

Prospect → Negotiation : Finalizing Installment Milestones

Prospect → Lost : Opted for Competitor

Negotiation → Converted : Placed with Partner Employer

Negotiation → Lost : Candidate Dropped Out

FollowUp → Contacted : Callback Completed

FollowUp → Interested : Positive Assessment

FollowUp → Lost : Unreachable After 5 Touches

Converted → [*] : Placed / Successfully Hired

NotInterested → [*] : Archived to Inactive Pool

Lost → [*] : Archived to Lost Pool

State Transition Rules Table

Current Stage	Allowed Next Stages	Preconditions & Validation Rules	Mandatory Reason Required
Cold Calling	Contacted , Not Interested , Lost	Must log call outcome or touchpoint.	Yes
New Lead	Contacted , Not Interested , Lost	Outreach touchpoint attempted.	Yes
Contacted	Interested , Follow-up , Not Interested , Lost	Counselor recorded initial interest score.	Yes
Interested	Prospect , Follow-up , Not Interested	Qualification confirmed, placement terms shared.	Yes
Prospect	Negotiation , Follow-up , Lost	Fee plan agreed; interview preparation initiated.	Yes
Negotiation	Converted , Lost	Interview cleared; corporate offer letter released.	Yes
Converted	None (Terminal)	Student accepted offer; payment milestone completed.	Yes

3.2 Placement Fee & Milestone Payment State Machine

stateDiagram-v2

[*] → Pending : Candidate Enrolled with Fee Plan

Pending → Overdue : Due Date Elapsed & Amount Unpaid

Pending → PartiallyPaid : Partial Transaction Recorded

Pending → FullyPaid : 100% Fee Received Upfront

Overdue → PartiallyPaid : Milestone Payment Received

Overdue → FullyPaid : All Dues Cleared

PartiallyPaid → Overdue : Subsequent Milestone Due Date Elapsed

PartiallyPaid → FullyPaid : Final Installment Paid

FullyPaid → [*] : Financial Realization Complete

4. Core Algorithms

4.1 Real-Time Financial Balance Calculation Algorithm

TYPESCRIPT

```
function recalculateCustomerBalance(customerId: string): void {
  const customer = StorageService.getById('crm_customers', customerId);
  const payments = StorageService.getData('crm_payments').filter(p => p.customerId === customerId);

  const totalPaid = payments.reduce((sum, p) => sum + Number(p.amount), 0);
  const pending = Math.max(0, Number(customer.totalFee) - totalPaid);

  customer.paidAmount = totalPaid;
  customer.pendingAmount = pending;

  const todayStr = new Date().toISOString().split('T')[0];
  const hasOverdueMilestone = Array.isArray(customer.installments) &&
    customer.installments.some(inst =>
      inst.status !== 'Paid' && inst.dueDate && inst.dueDate < todayStr
    );

  if (pending === 0 && customer.totalFee > 0) {
    customer.paymentStatus = 'Fully Paid';
  } else if (hasOverdueMilestone) {
    customer.paymentStatus = 'Overdue';
  } else if (totalPaid > 0) {
    customer.paymentStatus = 'Partially Paid';
  } else {
    customer.paymentStatus = 'Pending';
  }

  StorageService.update('crm_customers', customerId, customer);
}
```

4.2 Auto Round-Robin Lead Allocation Algorithm

TYPESCRIPT

```
function allocateLeadsRoundRobin(leads: RawLeadInput[], counselors: User[]): AssignedLead[] {
  if (!counselors.length) throw new Error("No active counselors available for assignment.");

  let currentIndex = 0;
  return leads.map(lead => {
    const counselor = counselors[currentIndex % counselors.length];
    currentIndex++;
    return {
      ...lead,
      salespersonId: counselor.id,
      salespersonName: counselor.name,
      managerId: counselor.managerId || null,
      managerName: counselor.managerName || null
    };
  });
}
```

Database Schema & Data Dictionary

[07_DATABASE_DESIGN_AND_DATA_DICTIONARY.md](#)

Database Design & Data Dictionary

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** DB-APEX-07
 - **Version:** 2.0
 - **Status:** Approved
 - **Target Database Engine:** PostgreSQL 16 Enterprise (ACID compliant)
-

1. Entity-Relationship Diagram (ERD)

```

erDiagram
    USERS ||--o{ CUSTOMERS : "assigned to"
    USERS ||--o{ USERS : "reports to"
    CUSTOMERS ||--o{ INSTALLMENTS : "scheduled under"
    CUSTOMERS ||--o{ PAYMENTS : "paid against"
    INSTALLMENTS ||--o{ PAYMENTS : "settles"
    CUSTOMERS ||--o{ STAGE_HISTORY : "tracks"
    CUSTOMERS ||--o{ CALLS : "logged on"
    CUSTOMERS ||--o{ FOLLOWUPS : "scheduled for"
    CUSTOMERS ||--o{ NOTES : "appended to"
    USERS ||--o{ ACTIVITIES : "triggered by"

    USERS {
        uuid id PK
        varchar name "NOT NULL"
        varchar email UK "NOT NULL"
        varchar password_hash "NOT NULL"
        varchar role "admin | manager | sales"
        uuid manager_id FK "REFERENCES USERS(id)"
        varchar phone
        varchar avatar_url
        boolean is_active "DEFAULT true"
        timestamp created_at "DEFAULT now()"
    }

    CUSTOMERS {
        uuid id PK
        varchar student_code UK "CRM-STU-XXXX"
        varchar name "NOT NULL"
        varchar mobile UK "10 digits, NOT NULL"
        varchar alt_mobile "WhatsApp / alternate"
        varchar email
        varchar qualification "Degree"
        varchar college "University"
        varchar passing_year "YOP: 2022-2026"
        varchar cgpa_percentage
        text skills
        varchar target_role
        varchar experience_level "Fresher | 0-1yr | 1-2yrs | 2+yrs"
        varchar expected_ctc
        varchar city
        varchar state
        varchar stage "Cold Calling | New Lead | Contacted | ..."
        varchar status "Active | Converted | Lost"
        varchar priority "High | Medium | Low"
        varchar source "Channel"
        uuid manager_id FK "REFERENCES USERS(id)"
        uuid salesperson_id FK "REFERENCES USERS(id)"
        decimal total_fee "CHECK (total_fee >= 0)"
        decimal paid_amount "CHECK (paid_amount >= 0)"
        decimal pending_amount "CHECK (pending_amount >= 0)"
        varchar payment_plan "Full Upfront | 2/3/4 Installments"
        varchar payment_status "Fully Paid | Partially Paid | Pending | Overdue"
        timestamp next_follow_up
        timestamp last_contacted
        timestamp created_at "DEFAULT now()"
        timestamp updated_at "DEFAULT now()"
    }
  
```

```

INSTALLMENTS {
  uuid id PK
  uuid customer_id FK "REFERENCES CUSTOMERS(id) ON DELETE CASCADE"
  int installment_number "1, 2, 3, 4"
  varchar label "e.g. Upfront Registration"
  decimal amount "NOT NULL"
  date due_date "NOT NULL"
  varchar status "Pending | Paid | Overdue"
  timestamp paid_date
  varchar transaction_ref
}

PAYMENTS {
  uuid id PK
  varchar receipt_number UK "REC-XXXXX"
  uuid customer_id FK "REFERENCES CUSTOMERS(id) ON DELETE CASCADE"
  uuid installment_id FK "REFERENCES INSTALLMENTS(id)"
  decimal amount "NOT NULL, CHECK (amount > 0)"
  varchar payment_mode "UPI | Net Banking | Card | Cash | Cheque"
  varchar transaction_ref "UTR / Cheque No."
  uuid received_by FK "REFERENCES USERS(id)"
  text notes
  timestamp payment_date "DEFAULT now()"
}

STAGE_HISTORY {
  uuid id PK
  uuid customer_id FK "REFERENCES CUSTOMERS(id) ON DELETE CASCADE"
  varchar from_stage "NOT NULL"
  varchar to_stage "NOT NULL"
  text reason "NOT NULL"
  uuid changed_by FK "REFERENCES USERS(id)"
  timestamp changed_at "DEFAULT now()"
}

CALLS {
  uuid id PK
  uuid customer_id FK "REFERENCES CUSTOMERS(id) ON DELETE CASCADE"
  uuid salesperson_id FK "REFERENCES USERS(id)"
  int duration_seconds "DEFAULT 0"
  varchar outcome "Connected | Busy | No Answer | ..."
  text notes
  timestamp called_at "DEFAULT now()"
}

FOLLOWUPS {
  uuid id PK
  uuid customer_id FK "REFERENCES CUSTOMERS(id) ON DELETE CASCADE"
  uuid salesperson_id FK "REFERENCES USERS(id)"
  date scheduled_date "NOT NULL"
  varchar scheduled_time "NOT NULL"
  varchar priority "High | Medium | Low"
  varchar type "Call | WhatsApp | Email | Meeting"
  text purpose "NOT NULL"
  varchar status "Pending | Completed | Cancelled"
  text outcome_notes
  timestamp completed_at
}

NOTES {
  uuid id PK
  uuid customer_id FK "REFERENCES CUSTOMERS(id) ON DELETE CASCADE"
  uuid author_id FK "REFERENCES USERS(id)"
  text content "NOT NULL"
}

```

```
        timestamp created_at "DEFAULT now()"
    }

    ACTIVITIES {
        uuid id PK
        uuid user_id FK "REFERENCES USERS(id)"
        uuid customer_id FK "REFERENCES CUSTOMERS(id) ON DELETE SET NULL"
        varchar action "NOT NULL"
        text description "NOT NULL"
        timestamp created_at "DEFAULT now()"
    }
```

2. Production PostgreSQL DDL Scripts

SQL

```
-- 1. USERS & ROLES TABLE
```

```
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(150) NOT NULL,
  email VARCHAR(255) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  role VARCHAR(30) NOT NULL CHECK (role IN ('admin', 'manager', 'sales')),
  manager_id UUID REFERENCES users(id) ON DELETE SET NULL,
  phone VARCHAR(20),
  avatar_url TEXT,
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

```
-- 2. CANDIDATE JOB SEEKERS TABLE
```

```
CREATE TABLE customers (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  student_code VARCHAR(30) UNIQUE NOT NULL,
  name VARCHAR(150) NOT NULL,
  mobile VARCHAR(15) UNIQUE NOT NULL,
  alt_mobile VARCHAR(15),
  email VARCHAR(255),
  qualification VARCHAR(100),
  college VARCHAR(255),
  passing_year VARCHAR(10),
  cgpa_percentage VARCHAR(20),
  skills TEXT,
  target_role VARCHAR(100),
  experience_level VARCHAR(30) DEFAULT 'Fresher',
  expected_ctc VARCHAR(50),
  city VARCHAR(100),
  state VARCHAR(100),
  stage VARCHAR(50) NOT NULL DEFAULT 'Cold Calling',
  status VARCHAR(50) NOT NULL DEFAULT 'Active',
  priority VARCHAR(20) NOT NULL DEFAULT 'Medium' CHECK (priority IN ('High', 'Medium', 'Low')),
  source VARCHAR(100) DEFAULT 'Website Inquiry',
  manager_id UUID REFERENCES users(id) ON DELETE SET NULL,
  salesperson_id UUID REFERENCES users(id) ON DELETE SET NULL,
  total_fee DECIMAL(12,2) NOT NULL DEFAULT 45000.00 CHECK (total_fee ≥ 0),
  paid_amount DECIMAL(12,2) NOT NULL DEFAULT 0.00 CHECK (paid_amount ≥ 0),
  pending_amount DECIMAL(12,2) NOT NULL DEFAULT 45000.00 CHECK (pending_amount ≥ 0),
  payment_plan VARCHAR(50) DEFAULT '2 Installments',
  payment_status VARCHAR(30) DEFAULT 'Pending' CHECK (payment_status IN ('Fully Paid', 'Partial')),
  next_follow_up TIMESTAMP WITH TIME ZONE,
  last_contacted TIMESTAMP WITH TIME ZONE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

```
-- 3. INSTALLMENT MILESTONES TABLE
```

```
CREATE TABLE installments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  customer_id UUID NOT NULL REFERENCES customers(id) ON DELETE CASCADE,
  installment_number INT NOT NULL,
  label VARCHAR(100) NOT NULL,
  amount DECIMAL(12,2) NOT NULL CHECK (amount > 0),
  due_date DATE NOT NULL,
  status VARCHAR(30) DEFAULT 'Pending' CHECK (status IN ('Pending', 'Paid', 'Overdue')),
  paid_date TIMESTAMP WITH TIME ZONE,
  transaction_ref VARCHAR(100)
);
```

-- 4. FINANCIAL PAYMENT TRANSACTIONS TABLE

```

CREATE TABLE payments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid()
  receipt_number VARCHAR(50) UNIQUE NOT NULL
  customer_id UUID NOT NULL REFERENCES customers(id) ON DELETE CASCADE
  installment_id UUID REFERENCES installments(id) ON DELETE SET NULL
  amount DECIMAL(12,2) NOT NULL CHECK (amount > 0)
  payment_mode VARCHAR(50) NOT NULL CHECK (payment_mode IN ('UPI', 'Net Banking', 'Credit/Debit'))
  transaction_ref VARCHAR(100) NOT NULL
  received_by UUID REFERENCES users(id) ON DELETE SET NULL
  notes TEXT
  payment_date TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- 5. STAGE TRANSITION AUDIT HISTORY

```

CREATE TABLE stage_history (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid()
  customer_id UUID NOT NULL REFERENCES customers(id) ON DELETE CASCADE
  from_stage VARCHAR(50) NOT NULL
  to_stage VARCHAR(50) NOT NULL
  reason TEXT NOT NULL
  changed_by UUID REFERENCES users(id) ON DELETE SET NULL
  changed_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- 6. TELEPHONY CALL LOGS

```

CREATE TABLE calls (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid()
  customer_id UUID NOT NULL REFERENCES customers(id) ON DELETE CASCADE
  salesperson_id UUID REFERENCES users(id) ON DELETE SET NULL
  duration_seconds INT DEFAULT 0
  outcome VARCHAR(100) NOT NULL
  notes TEXT
  called_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- 7. FOLLOW-UP TASKS

```

CREATE TABLE followups (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid()
  customer_id UUID NOT NULL REFERENCES customers(id) ON DELETE CASCADE
  salesperson_id UUID REFERENCES users(id) ON DELETE SET NULL
  scheduled_date DATE NOT NULL
  scheduled_time VARCHAR(20) NOT NULL
  priority VARCHAR(20) NOT NULL DEFAULT 'Medium'
  type VARCHAR(50) NOT NULL DEFAULT 'Call'
  purpose TEXT NOT NULL
  status VARCHAR(30) NOT NULL DEFAULT 'Pending' CHECK (status IN ('Pending', 'Completed', 'Cancelled'))
  outcome_notes TEXT
  completed_at TIMESTAMP WITH TIME ZONE
);

```

-- 8. COUNSELING NOTES

```

CREATE TABLE notes (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid()
  customer_id UUID NOT NULL REFERENCES customers(id) ON DELETE CASCADE
  author_id UUID REFERENCES users(id) ON DELETE SET NULL
  content TEXT NOT NULL
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

-- 9. AUDIT ACTIVITIES LOG

```

CREATE TABLE activities (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid()
  user_id UUID REFERENCES users(id) ON DELETE SET NULL
  customer_id UUID REFERENCES customers(id) ON DELETE SET NULL
  action VARCHAR(100) NOT NULL
  description TEXT NOT NULL
);

```

```
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
```

3. Indexing & Query Optimization Strategy

SQL

-- 1. Index on Candidate Lookups

```
CREATE INDEX idx_customers_salesperson ON customers(salesperson_id);
```

```
CREATE INDEX idx_customers_manager ON customers(manager_id);
```

```
CREATE INDEX idx_customers_stage ON customers(stage);
```

```
CREATE INDEX idx_customers_payment_status ON customers(payment_status);
```

```
CREATE INDEX idx_customers_passing_year ON customers(passing_year);
```

```
CREATE INDEX idx_customers_created_at ON customers(created_at);
```

-- 2. Composite Index for Excel Sheet Filtering

```
CREATE INDEX idx_customers_excel_composite ON customers(salesperson_id, stage, passing_year, payment_status);
```

-- 3. Financial Indexes

```
CREATE INDEX idx_payments_customer ON payments(customer_id);
```

```
CREATE INDEX idx_installments_due_date ON installments(due_date, status);
```

-- 4. Follow-up Index for Dashboard Alerts

```
CREATE INDEX idx_followups_schedule ON followups(salesperson_id, scheduled_date, status);
```

RESTful API Specification

`08_REST_API_SPECIFICATION.md`

RESTful API Specification & Contracts

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** API-APEX-08
- **Version:** 2.0
- **Base URL:** `https://api.crm.careerapex.com/api/v1`
- **Specification Standard:** OpenAPI 3.0 / RESTful JSON
- **Authentication:** HTTP Bearer Header (`Authorization: Bearer <JWT_TOKEN>`)

1. Global Standards & Error Envelope

1.1 Standard Response Format

Every API response returns a uniform JSON envelope:

JSON

```
"success" true
"statusCode" 200
"data": { }
"message" "Operation completed successfully."
"timestamp" "2026-09-30T14:30:00.000Z"
```

1.2 Standard Error Response Format

JSON

```
{
  "success": false,
  "statusCode": 400,
  "error": "BAD_REQUEST",
  "message": "Student primary mobile number is already registered.",
  "details": {
    "field": "mobile",
    "issue": "Duplicate phone number matched candidate CRM-STU-004."
  },
  "timestamp": "2026-09-30T14:30:00.000Z"
}
```

1.3 HTTP Status Codes

Code	Constant	Meaning
200	OK	Request succeeded; body contains data payload.
201	CREATED	Resource successfully created (Candidate, Payment, Follow-up).
400	BAD_REQUEST	Validation failed or business rule violation.
401	UNAUTHORIZED	Missing or invalid bearer JWT token.
403	FORBIDDEN	Role has insufficient permissions to perform action.
404	NOT_FOUND	Candidate or resource ID does not exist.
409	CONFLICT	Duplicate mobile phone number detected.
500	INTERNAL_SERVER_ERROR	Unhandled server exception.

2. API Endpoints Catalog

2.1 Authentication Endpoints

POST /auth/login

- **Access:** Public
- **Request Body:**

JSON

```
{
  "email": "raiesh.admin@careerapex.com",
  "password": "Password@123"
}
```

- **Response** 200 OK :

JSON

```
{
  "success": true,
  "data": {
    "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTJ5IiwiaWF0IjoxNTEyMzQ1NjQyfQ==",
    "expiresIn": 3600,
    "user": {
      "id": "usr-001",
      "name": "Raiesh Sharma",
      "email": "raiesh.admin@careerapiex.com",
      "role": "admin",
      "avatar": "../assets/avatars/admin.png"
    }
  }
}
```

2.2 Student Candidate Endpoints

GET /students

- **Access:** Authenticated (Data scoped automatically by user role)
- **Query Parameters:**
 - `search` (string): Text query across ID, Name, College, Skills, Mobile.
 - `stage` (string): Pipeline stage filter.
 - `paymentStatus` (string): `All` , `Fully Paid` , `Partially Paid` , `Pending` , `Overdue` .
 - `passingYear` (string): `2026` , `2025` , `2024` , etc.
 - `salespersonId` (string): Counselor ID filter.
 - `page` (integer): Page number (default: 1).
 - `pageSize` (integer): Rows per page (default: 10, max: 100).
- **Response** `200 OK` :

JSON

```
{
  "success": true,
  "data": {
    "items": [
      {
        "id": "CRM-STU-001",
        "name": "Aarav Sharma",
        "mobile": "9876543210",
        "email": "aarav.sharma@gmail.com",
        "qualification": "B.Tech Computer Science",
        "college": "IIT Delhi",
        "passingYear": "2025",
        "skills": "Java, Python, Spring Boot",
        "targetRole": "Full Stack Developer",
        "stage": "Converted",
        "status": "Placed / Hired",
        "priority": "High",
        "salespersonName": "Amit Verma",
        "totalFee": 45000,
        "paidAmount": 45000,
        "pendingAmount": 0,
        "paymentStatus": "Fully Paid"
      }
    ]
  },
  "pagination": {
    "page": 1,
    "pageSize": 10,
    "total": 24,
    "totalPages": 3,
    "hasNext": true,
    "hasPrev": false
  }
}
```

POST /students

- **Access:** Authenticated (Admin, Manager, Counselor)
- **Request Body:**

JSON

```
{
  "name": "Rohan Mehra",
  "mobile": "9845012345",
  "altMobile": "9845012346",
  "email": "rohan.m@gmail.com",
  "qualification": "B.E. Information Technology",
  "college": "BITS Pilani",
  "passingYear": "2025",
  "skills": "React, Node.js, PostgreSQL",
  "targetRole": "Frontend Engineer",
  "stage": "Cold Calling",
  "salespersonId": "usr-004",
  "totalFee": 45000,
  "paymentPlan": "2 Installments"
}
```

- **Response** `201 CREATED` : Returns created student candidate object.

POST /students/:id/stage

- **Access:** Authenticated (Scoped)

- **Request Body:**

JSON

```
{
  "newStage": "Interested",
  "reason": "Candidate cleared counseling screening and agreed to placement track."
}
```

- **Response** 200 OK : Updates candidate stage and logs to `crm_stage_history`.

2.3 Placement Fee & Payment Endpoints

POST /payments

- **Access:** Authenticated (Scoped)
- **Request Body:**

JSON

```
{
  "customerId": "CRM-STU-004",
  "installmentId": "inst-02",
  "amount": 22500,
  "paymentMode": "UPI",
  "transactionRef": "UPI/329482910384",
  "notes": "Paid via GooglePay to corporate bank account"
}
```

- **Response** 201 CREATED :

JSON

```
{
  "success": true,
  "data": {
    "receiptNumber": "REC-00124",
    "customerId": "CRM-STU-004",
    "amount": 22500,
    "paymentMode": "UPI",
    "transactionRef": "UPI/329482910384",
    "updatedBalance": {
      "totalFee": 45000,
      "paidAmount": 45000,
      "pendingAmount": 0,
      "paymentStatus": "Fully Paid"
    }
  }
}
```

GET /payments/receipt/:receiptNumber

- **Access:** Authenticated
- **Response** 200 OK : Returns printable receipt voucher payload including candidate details, amount in words, and authorized counselor signature.

2.4 Reports & Analytics Endpoints

GET /reports/revenue

- **Access:** Admin, Manager
- **Query Parameters:** startDate, endDate, counselorId, managerId, stage, paymentStatus.
- **Response** 200 OK :

JSON

```
{
  "success": true,
  "data": {
    "summary": {
      "totalBooked": 1600000,
      "totalCollected": 603500,
      "totalPending": 996500,
      "collectionRate": 37.7,
      "overdueCount": 6
    },
    "counselorLeaderboard": [
      {
        "counselorId": "usr-004",
        "name": "Amit Verma",
        "students": 8,
        "booked": 360000,
        "collected": 180000,
        "pending": 180000,
        "realizationRate": "50.0%"
      }
    ]
  }
}
```

Tech Stack & Migration Roadmap

09_TECH_STACK_AND_MIGRATION_ROADMAP.md

Technology Stack & Production Migration Roadmap

Career Apex CRM — Student Placement & Career Counseling Sales CRM

Document Control

- **Document Identifier:** TECH-APEX-09
- **Version:** 2.0
- **Status:** Approved
- **Audience:** CTO, Tech Leads, DevOps Engineers, Full-Stack Developers

1. Current Prototype vs. Production Technology Comparison

Dimension	Current Prototype Architecture	Target Enterprise Production Stack	Strategic Rationale
Frontend Framework	Pure HTML5, Modular Vanilla JS (ES6+), CSS3	Next.js 14 (App Router) + React 18 + TypeScript	Component reusability, strict compile-time typing, server-side rendering for reports.
Table Virtualization	Custom Vanilla DOM Pagination	TanStack Table v8 (Virtualizer)	Enables scrolling through 50,000+ candidates without DOM memory exhaustion.
Styling & Design System	Custom Vanilla CSS with CSS Variables	Tailwind CSS + Headless UI	Rapid component styling preserving custom CSS token hierarchy.
Backend API Engine	Client-Side In-Memory Controllers	Node.js (NestJS Framework) with TypeScript	Enterprise modular architecture, dependency injection, built-in validation pipes.
Database Persistence	Browser <code>localStorage</code> (JSON-serialized)	PostgreSQL 16 Enterprise	ACID financial guarantees for payment ledgers, relational integrity, row-level security.
Cache & In-Memory Store	In-memory JS variables	Redis 7 Cluster	Distributed session storage, API rate-limiting, and dashboard KPI caching.
Background Job Queues	Synchronous client execution	BullMQ + Redis Workers	Asynchronous bulk CSV ingestion without blocking HTTP request threads.
Document Storage	Base64 client strings	Cloudflare R2 / AWS S3	Secure storage for student resumes, identity proof, and PDF voucher receipts.
Telephony Integration	<code>tel:</code> URI Scheme	Exotel / Knowlarity / Twilio Cloud CTI	In-browser WebRTC softphone, automatic call recording, and exact duration logging.

2. Production Technology Stack Architecture

```

graph TD
    subgraph ClientLayer [Client Layer - Next.js 14 / TypeScript]
        ReactUI["Next.js React Client Components"]
        TanStack["TanStack Table Virtual Grid"]
        ChartUI["Chart.js / Recharts Visualizations"]
    end

    subgraph EdgeLayer [Edge & Security Layer]
        Cloudflare["Cloudflare CDN & DDoS Protection"]
        Nginx["Nginx Reverse Proxy / Load Balancer"]
    end

    subgraph BackendLayer [Backend Application Layer - NestJS Modular Monolith]
        APIGateway["NestJS API Gateway & Guards"]
        AuthModule["Auth & RBAC Module (Passport JWT)"]
        StudentModule["Student Candidate Module"]
        PaymentModule["Payment & Ledger Module"]
        ReportModule["Reporting & Analytics Engine"]
        QueueProducer["BullMQ Job Producer"]
    end

    subgraph WorkerLayer [Background Worker Layer]
        BullWorkers["BullMQ Consumer Workers (Node.js)"]
    end

    subgraph DataStorageLayer [Data & Storage Layer]
        Postgres["PostgreSQL 16 Master DB"]
        PostgresReplica["PostgreSQL Read Replica"]
        RedisCache["Redis 7 Cache & Queue Store"]
        S3Storage["AWS S3 / Cloudflare R2 Document Store"]
    end

    ClientLayer --> EdgeLayer --> BackendLayer
    BackendLayer --> RedisCache
    BackendLayer --> Postgres
    BackendLayer --> PostgresReplica
    BackendLayer --> QueueProducer --> RedisCache
    RedisCache --> BullWorkers --> Postgres
    BullWorkers --> S3Storage
  
```

3. Six-Phase Migration Roadmap (Prototype to Production)

```
gant
title Career Apex CRM Production Migration Roadmap (12 Weeks)
dateFormat YYYY-MM-DD
section Phase 1: Core Backend
Database Provisioning & Prisma ORM :2026-10-01, 10d
NestJS Architecture & JWT Authentication :2026-10-10, 11d
section Phase 2: Frontend API Integration
API Client Integration & State Management :2026-10-22, 14d
section Phase 3: Async Processing
BullMQ Bulk Ingestion & Deduplication Queue :2026-11-05, 14d
section Phase 4: Notifications
WhatsApp Cloud API & Email Reminders :2026-11-19, 14d
section Phase 5: Cloud Telephony
Cloud PBX WebRTC Softphone Integration :2026-12-03, 14d
section Phase 6: Deployment & QA
Security Audit, Docker & Production Launch :2026-12-17, 7d
```

Phase 1: Database Provisioning & NestJS API Foundation (Weeks 1–3)

- Provision managed PostgreSQL 16 cluster on AWS RDS / Supabase Enterprise.
- Execute SQL migrations for `users`, `customers`, `installments`, `payments`, `stage_history`, `calls`, `followups`, `notes`, and `activities`.
- Setup NestJS project structure with Prisma ORM.
- Implement JWT authentication with access/refresh tokens and RBAC middleware.

Phase 2: Frontend API Client Integration (Weeks 4–5)

- Create `api-client.ts` Axios adapter with request/response interceptors and automatic token refresh.
- Swap out `StorageService` calls with asynchronous REST endpoint calls.
- Maintain existing HTML5/CSS design system so visual components remain completely untouched.

Phase 3: High-Scale Bulk Processing & Worker Queues (Weeks 6–7)

- Configure Redis 7 cluster and initialize `BullMQ` worker pipeline.
- Implement background chunked ingestion for 100,000-row Excel spreadsheets.
- Add background duplicate phone scanning and counselor round-robin auto-distribution.

Phase 4: Production Notification Engine (Weeks 8–9)

- Integrate Meta WhatsApp Cloud API for automated candidate notifications (welcome message, fee reminders, interview schedule confirmations).
- Integrate AWS SES / SendGrid for transactional email dispatches.
- Implement Web Push Notifications for counselor follow-up alerts.

Phase 5: Cloud PBX Telephony Integration (Weeks 10–11)

- Integrate Exotel / Twilio WebRTC softphone inside the CRM layout.

- Implement automatic call duration tracking and secure S3 call recording storage.

Phase 6: QA, Penetration Testing & Production Deployment (Week 12)

- Docker containerization with multi-stage build optimization.
- GitHub Actions CI/CD pipeline targeting AWS ECS / Kubernetes.
- Vulnerability scanning, penetration testing, and load testing up to 10,000 concurrent requests.

Test Plan & Acceptance Test Cases (ATP)

[10_TEST_PLAN_AND_ACCEPTANCE_TEST_CASES.md](#)

Career Apex CRM — Comprehensive Test Plan & Acceptance Test Cases (ATP/QA)

Document Reference: SDLC-DOC-010**Version:** 1.0.0**Status:** Approved for QA Execution**Project:** Career Apex CRM (Student Job-Seeker Placement Platform)**Target Environment:** Staging / Production Pre-Flight

1. Executive Summary & Objectives

The purpose of this Test Plan is to define the testing strategy, test coverage, test environments, acceptance criteria, and specific executable test cases for the **Career Apex CRM** web application.

The primary objective is to guarantee:

1. Zero data leakage across role boundaries (strict RBAC enforcement).
2. 100% financial accuracy in placement fee, paid amount, and outstanding installment calculations.
3. High-throughput lead ingestion with zero duplicate candidate generation.
4. Seamless real-time updates across the Excel spreadsheet pipeline and management analytics dashboards.

2. Testing Scope & Methodology

2.1 In-Scope Functional Areas

- Authentication, Session Management & Role-Based Navigation Guards.
- Student Candidate Management (CRUD, education history, technical skills, YOP).
- Placement Fee Ledger (Total committed fee, multi-stage installment scheduling, payment recording, balance auto-recalculation).
- Excel Spreadsheet Pipeline View (Sticky columns/headers, quick stage switcher, mandatory disqualification/drop reason capture).
- Bulk Lead Ingestion (CSV/Excel parsing, cold-lead pipeline assignment, counselor round-robin allocation).
- Omnichannel Communications (WebRTC/Twilio click-to-call, WhatsApp deep-linking, mailto composition, call outcome logging).
- Follow-up Management (Scheduling, due date alerts, overdue status flags).

- Multi-Filter Analytics & Management Reporting (Date range, counselor, qualification, and stage cross-filtering).

2.2 Out-of-Scope (Deferred to Phase 2)

- Direct bank payment gateway reconciliation (Razorpay/Stripe automated webhooks).
- Native mobile application builds (iOS / Android wrappers).
- Automatic AI resume parsing from PDF/DOCX uploads.

3. Testing Levels & Strategy

```
graph TD
    subgraph "Testing Pyramid"
        E2E["End-to-End Testing (Playwright / Cypress)  
50+ Scenarios"]
        INT["Integration & API Testing (Supertest / Jest)  
180+ Test Cases"]
        UNIT["Unit Testing (Vitest / Jest)  
500+ Tests (Financial logic, deduplication, state machines)"]
    end
    UNIT --> INT --> E2E
```

Level	Tooling	Focus	Target Coverage
Unit Testing	Vitest / Jest	Pure functions, financial balance math, phone deduplication regex, pipeline state machines	> 90% Statement Coverage
Integration Testing	Supertest, Testcontainers (PostgreSQL)	REST API endpoints, JWT token guards, DB constraints, transactional rollback	> 85% API Endpoint Coverage
End-to-End (E2E)	Playwright	Full browser user journeys across Admin, Manager, and Counselor personas	100% Core Workflow Coverage
Performance / Load	k6	Bulk lead upload (50,000 rows), pipeline rendering (10,000 active students)	P95 latency < 500ms
Security / Pen-Test	OWASP ZAP, SonarQube	SQL injection, XSS sanitization, Broken Object-Level Authorization (BOLA)	Zero High/Critical Vulnerabilities

4. Test Environment & Test Data Setup

4.1 Environments

- **Local Dev:** Mocked `localStorage` / SQLite in-memory, hot-reload dev server.

- **Staging / QA:** Dockerized Next.js frontend, NestJS backend API, PostgreSQL 16 container, Redis 7 cache.
- **Mock Services:** Twilio sandbox voice endpoints, MailHog SMTP server for test transactional emails.

4.2 Seed Test Data Personas

- **Admin:** `admin@careerapex.com` / `AdminPass@2026`
- **Sales Manager:** `manager@careerapex.com` / `ManagerPass@2026`
- **Counselor 1 (Counselor A):** `sarah.j@careerapex.com` (assigned 50 candidates)
- **Counselor 2 (Counselor B):** `david.k@careerapex.com` (assigned 50 candidates)
- **Counselor 3 (Counselor C):** `priya.s@careerapex.com` (assigned 0 candidates)

5. Master Acceptance Test Matrix

Module 1: Authentication & Scoped Route Guards

Test ID	Scenario	Pre-Conditions	Action Steps	Expected Result	Priority
TC-01	Valid Counselor Login	User exists in DB with role <code>counselor</code>	1. Navigate to <code>/index.html</code> 2. Select Counselor role 3. Enter credentials 4. Click Login	Redirected to <code>/pages/salesperson-dashboard.html</code> . JWT token stored in secure cookie.	P1
TC-02	Valid Manager Login	User exists in DB with role <code>sales_manager</code>	1. Select Manager role 2. Enter valid credentials 3. Click Login	Redirected to <code>/pages/manager-dashboard.html</code> . Team oversight metrics displayed.	P1
TC-03	Valid Admin Login	User exists in DB with role <code>admin</code>	1. Select Admin role 2. Enter credentials 3. Click Login	Redirected to <code>/pages/admin-dashboard.html</code> . Full system navigation visible.	P1
TC-04	Invalid Password Attempt	User exists	1. Enter correct email, wrong password 2. Click Login	Red error banner displayed: "Invalid credentials". Session not created.	P2
TC-05	Unauthorized Route Access (Guard)	Logged in as Counselor	Attempt to navigate directly to <code>/pages/admin-dashboard.html</code> or <code>/pages/team.html</code>	Intercepted by route guard; redirected to <code>/pages/salesperson-dashboard.html</code> with warning toast "Access Denied".	P1

Module 2: Student Candidate Registration & Deduplication

Test ID	Scenario	Pre-Conditions	Action Steps	Expected Result	Priority
TC-06	Create New Candidate (Complete)	Logged in as Counselor	1. Navigate to Students page 2. Click "+ Add Student" 3. Enter Name, Email, 10-digit Phone, Degree (B.Tech CS), YOP (2025), Skills (Java, React) 4. Set Stage to "New Lead" 5. Click Save	Candidate created with unique ID. Appears at top of Students table. Toast confirms creation.	P1
TC-07	Phone Number Deduplication Block	Candidate with phone 9876543210 exists	1. Open "+ Add Student" 2. Enter different name/email, but same phone 9876543210 3. Click Save	Form validation stops submission. Alert shows: "Candidate with this phone number already assigned to Sarah Jenkins".	P1
TC-08	Phone Format Sanitization	Logged in	1. Input phone as +91 (987) 654-3210 2. Submit form	System strips special characters, normalizes to E.164 standard +919876543210, passes validation.	P2
TC-09	Mandatory Field Validation	Logged in	1. Click "+ Add Student" 2. Leave Name and Phone blank 3. Click Save	Input fields highlighted in red. Tooltips show "Full name and valid mobile number are required". Form does not submit.	P2
TC-10	Edit Student Candidate Profile	Candidate exists	1. Open Student Details 2. Click "Edit Profile" 3. Modify Degree to "MCA", Experience to "1 Year" 4. Click Save Changes	Details updated instantly in DB and UI. System activity logged: "Profile updated by Counselor A".	P2

Module 3: Placement Fee & Milestone Payment Ledger

Test ID	Scenario	Pre-Conditions	Action Steps	Expected Result	Priority
TC-11	Initialize Placement Package	Candidate created with no financial agreement	1. Open Student Details > Payment tab 2. Set Total Package Fee: ₹45,000 3. Plan: 3 Installments 4. Save agreement	Financial banner updates: Total Fee = ₹45,000, Paid = ₹0, Pending = ₹45,000. Payment Status = "Unpaid".	P1
TC-12	Record Partial Payment (Installment 1)	Package Fee = ₹45,000	1. Click "+ Record Payment" 2. Enter Amount: ₹15,000 3. Mode: UPI, Ref: UPI - 982348 4. Click Submit	Ledger adds ₹15,000 row. Paid Amount = ₹15,000, Balance = ₹30,000. Status updates to "Partial Paid". Activity logged.	P1
TC-13	Payment Exceeding Balance Block	Balance = ₹30,000	1. Click "+ Record Payment" 2. Enter Amount: ₹35,000 3. Click Submit	System raises error: "Payment amount (₹35,000) cannot exceed outstanding balance (₹30,000)". Form blocked.	P2
TC-14	Complete Full Payment (Settlement)	Balance = ₹30,000	1. Record payment of ₹30,000 2. Mode: Net Banking	Paid = ₹45,000, Balance = ₹0. Status badge switches to emerald green "Paid". Invoice download enabled.	P1
TC-15	Overdue Installment Notification Flag	Student has installment due date < Today and status = "Pending"	1. Load Follow-ups and Dashboard 2. Check overdue alerts	Student highlighted with amber warning icon: "Installment #2 Overdue by 4 days".	P2

Module 4: Excel Spreadsheet Pipeline & Stage Transitions

Test ID	Scenario	Pre-Conditions	Action Steps	Expected Result	Priority
TC-16	Excel Spreadsheet Layout Verification	100+ students in system	1. Navigate to <code>/pages/pipeline.html</code> 2. Verify DOM structure	Spreadsheet view displays with sticky headers, frozen checkbox and candidate columns, horizontal scroll for remaining columns.	P2
TC-17	Stage Change: Happy Path	Student in "Cold Calling"	1. Click stage dropdown in row 2. Change to "Connected / Interested"	Instant UI update. Toast displays: "Stage moved to Connected / Interested". Activity logged in candidate history.	P1
TC-18	Mandatory Reason on Stage "Not Interested"	Student in "Connected"	1. Change stage to "Not Interested"	Stage does not change immediately. Modal appears: "Reason for Drop". Options: Fee Issue, Opted Higher Studies, Invalid Contact.	P1
TC-19	Mandatory Reason Modal Cancellation	Modal open from TC-18	1. Click "Cancel" on drop modal	Modal closes. Candidate stage reverts to previous stage "Connected". No state change saved.	P2
TC-20	Mandatory Reason Form Submission	Modal open from TC-18	1. Select "Fee Issue" 2. Enter note: "Found ₹45k out of budget" 3. Click Confirm	Stage updates to "Not Interested". Reason badge saved and displayed in candidate profile timeline.	P1

Module 5: Bulk Excel/CSV Lead Ingestion & Allocation

Test ID	Scenario	Pre-Conditions	Action Steps	Expected Result	Priority
TC-21	Valid CSV Upload (Cold Calling Pipeline)	Manager logged in	1. Navigate to <code>/pages/import-leads.html</code> 2. Drag & drop <code>leads_batch_500.csv</code> 3. Select Target Pipeline: "Cold Calling" 4. Select Allocation: "Round-Robin to All Counselors" 5. Click Start Import	File parses without errors. Progress bar reaches 100%. 500 leads inserted. Distributed equally (~166 per counselor).	P1
TC-22	CSV with Missing Required Columns	File missing <code>phone</code> column	1. Upload <code>invalid_columns.csv</code>	File rejected at preview stage. Error table highlights: "Required column 'phone' is missing in header row".	P2
TC-23	Duplicate Phone Handling during Import	CSV contains 10 numbers already in CRM	1. Upload CSV with duplicate numbers 2. Run import	Summary report shows: "490 Leads Imported, 10 Duplicates Skipped". Downloadable error CSV provided with row details.	P1
TC-24	Targeted Counselor Batch Assignment	File with 100 leads	1. Target Pipeline: "Screening" 2. Counselor: "Sarah Jenkins" 3. Import	All 100 candidates assigned exclusively to Sarah Jenkins with stage "Screening".	P2
TC-25	Large File Ingestion Performance (5,000 rows)	Valid 5k CSV	1. Upload 5,000 lead file 2. Monitor memory and time	Import executes in asynchronous background chunks without browser freezing. Total ingestion time < 8 seconds.	P3

Module 6: Omnichannel Communication & Follow-up Scheduling

Test ID	Scenario	Pre-Conditions	Action Steps	Expected Result	Priority
TC-26	Click-to-Call Modal & Auto-Log	Counselor logged in	1. Open student row or profile 2. Click green phone icon	Call modal opens displaying student name and number. Call duration timer starts. Audio connects.	P2
TC-27	Call Outcome Logging & Next Follow-up	Active call finished	1. Hang up call 2. Select Outcome: "Busy / Callback Requested" 3. Pick Next Follow-up: Tomorrow 11:00 AM 4. Save Log	Call record saved with duration and outcome. New follow-up scheduled and visible on <code>/pages/followups.html</code> .	P1
TC-28	WhatsApp Web Link Trigger	Valid phone number	1. Click WhatsApp icon in candidate row	New browser tab opens with URL: <code>https://wa.me/919876543210?text=Hi%20...</code> . Activity logged: "WhatsApp conversation initiated".	P2
TC-29	Follow-up Status Completion	Follow-up due today	1. Navigate to <code>/pages/followups.html</code> 2. Check box "Mark as Completed" 3. Enter resolution note	Follow-up moves from "Pending" to "Completed" tab. Candidate last contacted timestamp updated.	P2
TC-30	Overdue Follow-up Badge Calculation	Follow-up date was yesterday	1. View Counselor Dashboard	"Overdue Follow-ups" widget counter increments by 1. Highlighted in red font.	P2

Module 7: Universal Multi-Filter Intelligence Reports

Test ID	Scenario	Pre-Conditions	Action Steps	Expected Result	Priority
TC-31	Date Range Filter Application	Manager logged in	1. Navigate to <code>/pages/reports.html</code> 2. Select Date: "This Month" 3. Click Apply	All 4 KPI summary cards (Leads, Revenue, Conversion Rate, Placements) recalculate dynamically for the selected month.	P1
TC-32	Counselor Breakdown Filter	Multiple counselors active	1. Select Counselor: "David Kim" 2. Click Apply	Stage funnel chart and conversion rates update to reflect only David Kim's pipeline candidates.	P1
TC-33	Degree / Qualification Cross-Filter	Diverse student pool	1. Select Qualification: "B.Tech CS" 2. Select Stage: "Interview Scheduled"	Table filters to show only Computer Science students currently attending job interviews.	P2
TC-34	Export Filtered Report to CSV	Filter applied	1. Click "Export to CSV" button	File <code>career_apex_report_[timestamp].csv</code> downloads with exact filtered row dataset matching the current view.	P2
TC-35	Reset All Filters	Multiple filters active	1. Click "Reset Filters"	All filter dropdowns revert to "All", date resets to current 30-day window, KPIs and charts reload full dataset.	P2

6. Defect Severity & Priority Classification

graph TD
 P1["P1 — Blocker / Critical
Security breach, data loss, calculation bug, app crash"]
 P2["P2 — Major
Primary workflow broken, no workaround available"]
 P3["P3 — Moderate
Minor feature defect, cosmetic layout glitch, workaround exists"]
 P4["P4 — Trivial / Enhancement
Spelling error, minor color misalignment"]

SLA for Defect Resolution:

- **P1 Blocker:** Immediate hotfix within 4 hours. QA validation required before deployment.
- **P2 Major:** Fix within 24 hours. Included in daily staging regression run.
- **P3 Moderate:** Scheduled for current sprint milestone.
- **P4 Trivial:** Backlog prioritization for UI polish cycles.

7. QA Sign-Off & Exit Criteria

The Career Apex CRM shall be certified as **Production-Ready** only when all the following quantitative criteria are satisfied:

1. **100% Execution** of all 35 Acceptance Test Cases (TC-01 through TC-35).
 2. **Zero Open P1 or P2 Defects** in the issue tracking system.
 3. **P3/P4 Defect Count** does not exceed 5 non-critical issues.
 4. **Performance Threshold:** Pipeline sheet load time under 1.5 seconds for 5,000 active records on standard broadband.
 5. **Security Clearance:** 100% pass on OWASP Top 10 automated scans with zero high/critical vulnerabilities.
 6. **Formal Sign-off:** Approved by Lead QA Engineer, Solutions Architect, and Lead Product Owner.
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End of Test Plan & Acceptance Test Cases Document.

